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Adversarial robustness of variable-star classifiers within the photometric error budget

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ABSTRACT

Context. Machine-learned classifiers reach excellent accuracy in variable-star catalogs, but their stability under perturbations within the reported photometric uncertainties is unknown.
Aims. We introduce a photometrically bounded threat model in which each magnitude may be displaced only within its reported photometric error interval, quantify the vulnerability of standard light-curve classifiers, and determine what closes this scientific gap.
Methods. We craft adversarial light curves that shift each observed magnitude only inside its reported error interval, using white-box projected gradient descent and query-only black-box searches. As a defense we train a two-tower convolutional encoder on the raw differential representation $(\Delta t, \Delta m, \sigma_{\Delta m})$ with adversarial training, and audit it with adaptive white-box, gradient-free, margin-based, and attack-potency evaluations on OGLE-III, *Gaia* DR2, and WISE.
Results. Standard classifiers are vulnerable: a convolutional neural network (CNN) falls from 96.4% to 65.8% on OGLE-III, and the 16-feature random forest loses 6.5 percentage points (pp) at half the error budget under a feature-recomputing black-box attack. Across surveys, plain adversarial training is the operative defense — additional consistency, contrastive, and information-bottleneck terms never improve on it — and the raw differential representation is associated with a further 3–7 pp advantage over an adversarially trained CNN. The defended encoder loses at most 1.0 pp under adaptive attack (97.3% on OGLE-III, 90.4% on *Gaia*, 92.5% on WISE, where two under-represented classes limit balanced accuracy to 61.0%), and vulnerability scales with the error budget: *Gaia*, with the smallest uncertainties, is nearly immune even undefended. Prediction changes concentrate in $\lesssim 0.5\%$ of sources, providing a per-source robustness flag.
Conclusions. Photometric uncertainty defines a physically motivated attack surface, and worst-case-within-errors evaluation is a practical catalog-auditing tool. Adversarial training on a raw differential representation closes the vulnerability at a clean-accuracy cost of ∼0.5 pp and survives adaptive, gradient-free, and margin-based attacks; the apparent robustness of feature-based methods does not survive attacks tailored to them.

Key words. stars: variables: general – methods: data analysis – methods: statistical – techniques: photometric

1. Introduction

There have been considerable efforts to automatically classify and extract knowledge from variable star catalogs in recent decades. This focus is primarily motivated by the significance of certain stars (e.g., RR Lyrae and Cepheids), which are critical for measuring cosmological distances (**??**). In this context, machine learning models, particularly deep learning approaches, have become crucial in automatically classifying variable stars (Aguirre et al. 2019). Despite their success, a gap remains in our understanding of the robustness of these models in complex scenarios, such as biased data and low signal-to-noise ratios.

Robustness and uncertainty are closely linked: a robust model must remain stable on noisy or uncertain data. The photometric error is the natural quantification of uncertainty in a light curve, yet while some feature-based classifiers and a few deep models incorporate it as an input or weighting (e.g., Naul et al. 2018), none has studied the *vulnerability* of the prediction to perturbations contained within it. If a classifier changes its answer under perturbations smaller than the reported measurement uncertainty, its output for that star reflects the particular noise realization that was recorded rather than a property of the star itself.

In this paper, we examine the robustness of these models by assessing their vulnerability to small perturbations constrained by the reported photometric errors. Although these perturbations are bounded by the observational uncertainties, they can significantly alter model predictions, exposing the fragility of standard classifiers. Our contributions are threefold: (i) a photometrically bounded threat model and attack protocol in which every magnitude moves by at most a fixed multiple of its reported uncertainty, together with an auditing methodology (adaptive, gradient-free, and margin-based evaluations, and a per-source robustness flag); (ii) the demonstration, via controlled ablations and an adversarially trained CNN baseline, that standard adversarial training transfers to irregularly sampled light curves at a clean-accuracy cost of only ∼0.5 percentage points and is the operative defense; and (iii) a robust classifier that combines this training with the raw differential representation of the light

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curve, which contributes additional clean and adversarial accuracy beyond what adversarial training alone provides.

Worst-case sensitivity is the subject of adversarial-example research: imperceptible input modifications that fool a classifier (Goodfellow et al. 2015; Papernot et al. 2016b), and that transfer between models trained on the same task (Papernot et al. 2016a), enabling benchmark-style robustness comparisons. Adversarial examples are well studied for images but remain underexplored for irregularly sampled time series and no previous work ties the size of the allowed perturbation to the reported uncertainty of each individual measuremment. Figure 1 shows a light-curve example: the two curves are indistinguishable within their error bars, yet a standard convolutional classifier assigns them to different classes.

Robustness to adversarial examples also carries collateral benefits — adversarial training can reduce overfitting (Miyato et al. 2015) and adversarial examples retain class-relevant information useful under imbalance (Ilyas et al. 2019) – and in astronomy, observational effects such as noise or PSF blurring can act as natural adversarial perturbations (Ćiprijanović et al. 2022).

Defense methods either inject adversarial examples into training — the minimax adversarial training of Madry et al. (2018), extended by the clean–adversarial consistency term of Zhang et al. (2019) — or denoise the inputs with generative models (Samangouei et al. 2018; Li & Ji 2019). These strategies have been developed almost exclusively for images, and their transfer to unevenly sampled astronomical time series has not been tested.

In this paper, we examine the robustness of variable-star classifiers by assessing their vulnerability to small perturbations constrained by the reported photometric errors: every magnitude is allowed to move by at most a fixed multiple of its reported uncertainty. We first demonstrate the vulnerability of traditional classifiers — Random Forest (RF), convolutional neural network (CNN), and recurrent neural network (RNN) baselines — on data from the Optical Gravitational Lensing Experiment (OGLE-III; Udalski 2004), *Gaia* Data Release 2 (DR2; Gaia Collaboration et al. 2018), and the Wide-field Infrared Survey Explorer (WISE; Wright et al. 2010), and show that the apparent robustness of feature-based classifiers degrades when the attack targets them directly. We then present a supervised classifier that operates directly on the raw differential representation of the light curve — per-point time differences Δt, magnitude differences Δm, and their propagated photometric uncertainties $\sigma_{\Delta m}$ — and is explicitly trained to be robust to perturbations within the photometric error, and we use controlled ablations to establish which components of the training objective are responsible for that robustness. Finally, we audit the resulting model with adaptive, gradient-free, and margin-based attacks. To our knowledge, this is among the first studies of photometrically bounded adversarial perturbations in variable-star classification operating directly on the observed time series.

This paper is organized as follows. In Sect. 2, we review previous work on variable star classification and adversarial learning. Section 3 details the proposed model, and Sect. 4 describes the data sets used. Section 5 presents our results, Sect. 6 discusses their implications and limitations, and Sect. 7 concludes.

2. Related work

2.1. Variable star classifiers

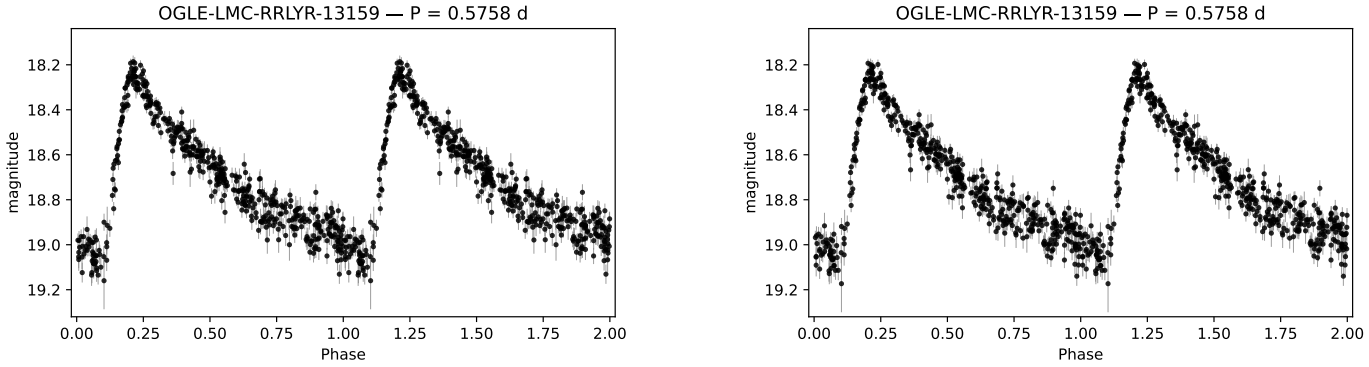
Machine-learned classification of variable stars was first built on hand-engineered features. Debosscher et al. (2007) pioneered the approach with 28 photometric features on OGLE and MACHO data; Richards et al. (2011) were the first to incorporate the photometric error explicitly (through the `beyond1std` feature), and the FATS library (Nun et al. 2015) consolidated over 65 features, seven of which are error-aware. Subsequent work used error-aware features to suppress spurious variability detections (Sokolovsky et al. 2017; Pashchenko et al. 2018) and modeled sparse light curves probabilistically with Gaussian processes (Castro et al. 2017).

Deep classifiers later replaced feature extraction with end-to-end learning on the light curve itself, and several explicitly use the photometric error: the autoencoding RNN of Naul et al. (2018) weights its loss by the heteroskedastic uncertainties, an idea extended with Gaussian-mixture modeling by Tsang & Schultz (2019); Hosenie et al. (2020) augment minority classes with Gaussian-process resampling of the errors; Donoso-Oliva et al. (2021) feed the raw triplets of observation time t, magnitude m, and photometric error e directly to (phased) long short-term memory networks (LSTMs); and Kang et al. (2023) scale synthetic uncertainties to match real ones in an RNN–CNN hybrid. In all of these works the error improves average accuracy or tolerance to noise; none evaluates stability under worst-case perturbations within the error bars.

The current state of the art is dominated by transformer-based and self-supervised representations. Donoso-Oliva et al. (2023) propose ASTROMER, they pre-train a transformer encoder on millions of unlabeled curves, and its successor ASTROMER 2 (Donoso-Oliva et al. 2026) weights the pre-training loss by the reciprocal photometric uncertainties. Cabrera-Vives et al. (2024) combine light-curve and tabular transformers for alert streams. Contrastive pre-training has also been applied to large photometric archives (Ding et al. 2026), a mechanism related to the supervised contrastive term of our objective (Sect. 3.2.4). Closest to our concerns, Cádiz-Leyton et al. (2025) equipped ASTROMER-based classifiers with predictive-uncertainty estimates. All of these models, however, are trained and evaluated on clean inputs. A complementary line of work addresses reliability under training-set bias and distribution shift for RR Lyrae classifiers (Pérez-Galarce et al. 2021, 2023, 2025); that notion of robustness operates at the population level, whereas the instance-level stability studied here — the worst-case sensitivity of a single prediction within the reported error bars — had not been evaluated for variable-star classifiers prior to this work.

2.2. Adversarial learning

Adversarial examples — small input perturbations that flip a model's prediction — were first demonstrated for deep networks by Szegedy et al. (2013) and attributed to their locally linear behavior by Goodfellow et al. (2015). They transfer across architectures and training sets (Papernot et al. 2016a; Liu et al. 2016; Dong et al. 2018), which enables black-box attacks. For time series, Ismail Fawaz et al. (2019a) showed that small perturbations fool standard deep classifiers for univariate series, and Karim et al. (2020) extended the result to multivariate series with a distilled-network attack; the architectures they attack are the standard CNN/RNN families reviewed by Ismail Fawaz et al.



(a) Original light curve

(b) Adversarial light curve

Fig. 1: Original and adversarial versions of the same OGLE RR Lyrae source (OGLE-LMC-RRLYR-13159), phase-folded with its 0.576-day period and shown with the reported per-point photometric uncertainties $\pm e_i$ as error bars. Panel (a) shows the clean light curve, and panel (b) shows the adversarial version drawn from the adversarial test set of Section 3.1.2: each observed magnitude is displaced within the pointwise bound $|m_i^{\mathrm{adv}} - m_i| \leq k\, e_i$ at the default budget $k = 0.5$. By construction, every displaced magnitude remains inside its error interval — the two panels are visually almost indistinguishable — yet the bounded displacement changes the CNN prediction from RR Lyrae to Cepheid.

(2019b). Pialla et al. (2022, 2023) showed that adding a smoothness penalty to iterative time-series attacks yields perturbations that are harder to detect at no cost in effectiveness — the same attack family that our smoothness-regularized projected gradient descent PGD (Sect. 3.1.2) and its detectability analysis (Appendix A) build on.

In astronomy, adversarial fragility has been documented in several data modalities: Gebhard et al. (2019) exhibited adversarial failure modes of convolutional gravitational-wave detectors; Campagne (2020) applied adversarial training to photometric-redshift networks; Ćiprijanović et al. (2022) found that observational noise can act as a natural adversarial example for galaxy-morphology classifiers trained on simulated images representative of the Legacy Survey of Space and Time (LSST) at the Vera C. Rubin Observatory (Ivezić et al. 2019); Horowitz & Melchior (2022) biased direct parameter-inference networks with small-amplitude systematics; and Jadhav et al. (2023) used adversarial attacks to expose, and then retrain away, failure modes of gravitational-wave detection models. **?** also examined adversarial robustness in CNN-based classification of gravitational-wave glitches using time–frequency images. For variable-star catalogs, the relevant question is whether predictions remain stable under perturbations of irregularly sampled light curves, with a perturbation budget defined by the reported photometric uncertainty of each observation. None of these works concerns irregularly sampled photometric time series, and none ties the perturbation budget to the reported per-point measurement uncertainty, as we do here.

On the defense side, adversarial training (Madry et al. 2018) remains the standard: the model is trained on worst-case perturbations within a bounded set. Tradeoff-inspired adversarial defense via surrogate-loss minimization (TRADES; Zhang et al. 2019) adds an explicit consistency term between clean and adversarial predictive distributions; supervised contrastive learning (Khosla et al. 2020) organizes the latent space by class and was extended to adversarial views by Kim et al. (2020); and the variational information bottleneck (VIB) (Alemi et al. 2017) limits the input–latent mutual information, filtering out non-informative perturbations. A complementary, certified alternative is randomized smoothing (Cohen et al. 2019), which is natural to mention in this context because Gaussian measurement noise has a direct physical meaning for photometric data; we focus on empirical adversarial training and leave certified defenses to future work. These strategies have been developed almost exclusively for images; their application to unevenly sampled astronomical time series — both attack methodology and defense design — remains essentially unexplored and is the gap this paper addresses.

3. Method description

The framework comprises two stages. In Stage 1 we generate photometrically bounded adversarial light curves by training a surrogate CNN in a two-channel $(\Delta t, \Delta m)$ representation and applying a PGD attack constrained by the per-point photometric uncertainty. In Stage 2 we build a supervised classifier that operates directly on the raw differential representation of the light curves and is trained adversarially — with three optional representation-shaping regularizers – so that its predictions remain stable under the perturbations generated in Stage 1. An overview of the entire pipeline is shown in Fig. E.1 (Appendix E).

3.1. Stage 1: Photometry-constrained adversarial light-curve generation

3.1.1. Surrogate CNN in $(\Delta t, \Delta m)$ space

We first train a CNN following the two-tower design of Aguirre et al. (2019). Each light curve is represented by two aligned one-dimensional sequences of length N: the time differences $\Delta t = (t_2 - t_1, t_3 - t_2, \ldots)$ and the magnitude differences $\Delta m = (m_2 - m_1, m_3 - m_2, \ldots)$. The CNN processes each channel with shared one-dimensional convolutional blocks, concatenates the resulting feature maps, and outputs class probabilities via a softmax layer. Once trained, the surrogate provides access to the input gradients $\nabla_{\mathbf{x}} \mathcal{L}(h_{\mathrm{cnn}}(\mathbf{x}), y)$, where $\mathbf{x} = (\Delta t, \Delta m)$ and $\mathcal{L}$ is the cross-entropy loss. These gradients are the building block of the adversarial attack described next.

3.1.2. Iterative PGD attack with magnitude-level hard projection

Our goal is to craft adversarial curves that fool the classifier while remaining *photometrically bounded*, i.e. admissible under the pointwise photometric-error constraint: every perturbed magnitude must stay within a fraction of its reported observational uncertainty. Concretely, for an observed light curve $\{(t_i, m_i, e_i)\}_{i=1}^N$ with per-observation photometric uncertainties e_i, we seek an adversarial counterpart $\{(t_i, m_i^{\rm adv}, e_i)\}_{i=1}^N$ that shares the original observation times t_i and uncertainties e_i and modifies only the magnitudes within the box constraint

$$|m_i^{\rm adv} - m_i| \;\le\; k\, e_i, \qquad i = 1, \ldots, N, \tag{1}$$

where $k \ge 0$ is a real-valued, dimensionless budget parameter. For this threat model, we restrict k to $[0, 1]$, using $k = 0.5$ by default and $k = 1$ for evaluations at the full reported uncertainty. Coordinate-dependent budgets of this kind have precedent in tabular adversarial machine learning, where per-feature perturbation bounds encode feature semantics (Ballet et al. 2019; Erdemir et al. 2021); here the per-point bound is set by the measurement uncertainty of each observation.

Because the surrogate CNN of Section 3.1.1 consumes first differences rather than raw magnitudes, the attacked tensor must eventually be expressed in $(\Delta t, \Delta m)$ form. We nevertheless parameterize the attack in the same space in which the physical constraint of Eq. (1) is defined: the per-observation magnitude displacement $\delta_i = m_i^{\rm adv} - m_i$. The first observation is kept fixed ($\delta_1 = 0$), which removes an arbitrary global magnitude offset that would disappear after differencing. We note that, because the budgets are heteroscedastic, this anchoring is not an exact gauge choice: it explores a strict subset of the set defined by Eq. (1), since a feasible $\boldsymbol{\delta}$ shifted by $-\delta_1$ need not satisfy every pointwise bound. We verified empirically that the restriction is immaterial: re-running the adaptive white-box attack of Sect. 5.7 on OGLE-III with δ_1 free within its own budget $k\, e_1$ leaves the adversarial accuracy of Robust Raw Encoder (RRE) unchanged at 0.973 (identical to four decimal places), as expected when one observation out of ∼500 is released. For a candidate displacement vector $\boldsymbol{\delta}$, the magnitude-difference channel evaluated by the CNN is

$$\Delta m_i(\boldsymbol{\delta}) = \Delta m_i + \delta_i - \delta_{i-1}, \qquad i \ge 2, \tag{2}$$

while the cadence channel Δt is never modified.

Magnitude-space PGD update. At iteration r, PGD updates the raw-magnitude displacement and immediately projects it back to the photometric box:

$$\boldsymbol{\delta}^{(r+1)} \leftarrow \Pi_{\mathbf{e},k}\Big[\boldsymbol{\delta}^{(r)} + \boldsymbol{\alpha} \odot \operatorname{sign}\Big(\nabla_{\boldsymbol{\delta}}\, \mathcal{L}\Big(h_{\rm cnn}(\Delta t, \Delta m(\boldsymbol{\delta}^{(r)})), y\Big)\Big)\Big], \tag{3}$$

where $\odot$ denotes element-wise multiplication (the Hadamard product), and $\Pi_{\mathbf{e},k}$ clips each component to $[-ke_i, +ke_i]$ and keeps $\delta_1 = 0$. Thus every iterate, and not only the final adversarial curve, satisfies Eq. (1). The corresponding heteroscedastic uncertainty used as an input channel by the defended model is the standard propagated error of the first difference,

$$\sigma_{\Delta m_i} = \sqrt{e_i^2 + e_{i-1}^2}. \tag{4}$$

The triple $\{(t_i, m_i + \delta_i, e_i)\}_{i=1}^N$ is the final adversarial light curve: it shares the times and reported uncertainties of the original observation, and its magnitudes differ from the clean values by at most $k\, e_i$. When this curve is later fed into any classifier that operates on a differenced representation – including the surrogate CNN and our own robust classifier (Section 3.2) – the differential sequences are recomputed from these bounded magnitudes and the original observation times, as described in Section 3.2.1.

Hyperparameters and smoothness regularization. We use $k = 0.5$ throughout this work, $T = 20$ PGD iterations, three random restarts, step size $\alpha_i = 0.2\, k\, e_i$, and a uniform random initialization within the photometric box (Madry et al. 2018). We adopt $k = 0.5$ as the default operating point because it probes vulnerability well inside the admissible set – each point moves by at most half its reported 1σ uncertainty – while the maximal budget considered, $k = 1.0$, reported alongside, saturates the error bars. The inner loop includes a small temporal smoothness penalty $\lambda_{\rm sm} = 10^{-3}$ on the second discrete differences of $\boldsymbol{\delta}$, following the smooth time-series attack family of Pialla et al. (2022). This term is *not* meant to make the perturbation mimic photometric noise: while the white-noise component of the photometric error is temporally uncorrelated, correlated effects – red noise, blending, detrending residuals and other systematics of survey photometry – do produce temporally structured deviations, and a smooth bounded perturbation is in fact the admissible-set member closest to that family; the penalty itself only steers PGD away from high-frequency, low-gradient solutions. The resulting displacements saturate the pointwise budget but are temporally correlated: Appendix A characterizes this structure and shows that order-sensitive statistics (von Neumann ratio, lag-one autocorrelation) expose the attack while amplitude-based statistics do not.

3.1.3. Transferability across architectures

Adversarial examples generated against one classifier often transfer to others trained on the same task (Papernot et al. 2016a). To investigate this property in our setting, we evaluated RF, CNN, and RNN on the same set of adversarial light curves generated from the surrogate CNN of Section 3.1.1, allowing a controlled comparison of the robustness of each architecture under a single shared attack.

3.2. Stage 2: Robust supervised classifier on raw light curves

3.2.1. Input representation: raw differential sequences

Our classifier consumes the light curves directly in the time domain. For every light curve $\{(t_i, m_i, e_i)\}_{i=1}^N$ we build three aligned sequences of consecutive differences: the time gaps $\Delta t_i = t_i - t_{i-1}$, the magnitude differences $\Delta m_i = m_i - m_{i-1}$, and their propagated uncertainties $\sigma_{\Delta m_i} = \sqrt{e_i^2 + e_{i-1}^2}$ (Eq. (4)). Differencing makes the representation invariant to the physically uninformative zero-point of the magnitude scale and to the epoch of the first observation, while retaining the local shape and cadence of the variability. It is the only representation our classifier uses.

Fixed-length sequences and padding. Light curves have a heterogeneous number of observations. We adopted a fixed length of $L = 500$ points to retain more of the available observations for classification and adversarial evaluation, particularly in OGLE-III. This sequence length also has precedent in the variable-star classifier of Aguirre et al. (2019). Curves with more than L epochs are uniformly subsampled in index along

their full time baseline – preserving the native ordering and the long-term cadence rather than discarding the later epochs – and shorter curves are right-padded. Padded positions carry a zero raw difference in every channel, which after the global standardization described next maps to the constant $-\mu/\sigma$ of each channel; the convolutions are not masked, so these constant tails are processed like ordinary input. We consider two aspects of unmasked padding: its interaction with the adversarial attack and its potential effect on classification performance. First, the attack cannot act on the padding: the budget encoding marks padded positions as non-perturbable, so all adversarial displacement is confined to real observations. Second, although global average pooling includes the padded tail – a potential dilution of the signal for short curves, most acute for *Gaia* DR2, whose ∼20–30 epochs occupy < 6% of the tensor – the effect is not visible empirically, the clean accuracy of RRE on *Gaia* DR2 is flat with curve length (0.92 below 15 epochs, 0.89–0.92 between 15 and 40, and 0.91 above 40), with no monotonic degradation as the padding fraction grows.

Global per-channel standardization. Each channel is standardized with a single mean and standard deviation computed once on the training split and reused for the validation and test splits and for the adversarial counterparts. This keeps clean and adversarial inputs on a common, fixed scale and avoids any per-curve quantity that could itself depend on the class.

The encoder receives two input branches (Sect. 3.2.2): a *cadence channel* $\mathbf{x}_t = \Delta t \in \mathbb{R}^{L\times 1}$ and a *magnitude channel* $\mathbf{x}_m = (\Delta m, \sigma_{\Delta m}) \in \mathbb{R}^{L\times 2}$. The uncertainty sub-channel lets the encoder weight noisy points differently from precise ones, and keeps the differential input self-consistent, since the uncertainty of Δm_i must include the independent errors of both measurements. The adversarial budget is enforced separately in raw-magnitude space through e_i, so $\sigma_{\Delta m}$ is never modified.

3.2.2. Encoder architecture with variational bottleneck

The classifier is built around a two-tower one-dimensional convolutional encoder following the two-channel CNN of Aguirre et al. (2019) used for the Stage 1 surrogate: one tower for $\mathbf{x}_t$ and one for $\mathbf{x}_m$, fused before a joint convolutional block.

Two-tower convolutional backbone. Each tower applies two ReLU convolutional blocks (Conv1D 32×7, then 64×5 with stride 2); the feature maps are concatenated and processed by two joint blocks (192×5, stride 2; 256×3), followed by global average pooling and a dense layer of width 256, whose output we denote $\mathbf{h}_\theta(\mathbf{x}) \in \mathbb{R}^{256}$, with $\mathbf{x} = (\mathbf{x}_t, \mathbf{x}_m)$.

Variational bottleneck. Following the variational autoencoder construction of Kingma & Welling (2014), two linear heads on $\mathbf{h}_\theta(\mathbf{x})$ parameterize a Gaussian posterior $q_\theta(\mathbf{z} \mid \mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}_\theta(\mathbf{x}), \mathrm{diag}\,\boldsymbol{\sigma}^2_\theta(\mathbf{x}))$ on a $D = 48$ dimensional latent space. During training a latent sample is drawn with the reparameterization trick, $\mathbf{z} = \boldsymbol{\mu}_\theta + \boldsymbol{\sigma}_\theta \odot \boldsymbol{\epsilon}$ with $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$; at inference we use $\mathbf{z} = \boldsymbol{\mu}_\theta(\mathbf{x})$. This is the stochastic encoder of the VIB (Alemi et al. 2017), whose Kullback–Leibler (KL) term enters the training objective (Sect. 3.2.4).

Classifier and projection heads. The latent vector feeds a *classifier head* π_ϕ (Dense(64) → Dropout(0.2) → Dense(4) → Softmax), whose output $\mathbf{p} = \pi_\phi(\mathbf{z})$ enters the cross-entropy and TRADES terms and gives the predicted class, and an ℓ_2-normalized *projection head* g_ψ (Dense(128) → Dense(64)), used only by the contrastive term and discarded after training. All weights are trained end-to-end from scratch.

3.2.3. Inner adversarial attack during training

We follow the adversarial-training minimax recipe of Madry et al. (2018): at every training step a worst-case counterpart of each clean curve is generated against the current model, and the parameters are updated to minimize the loss on both. The inner maximization is a short PGD attack parameterized, as in Stage 1, in the space where the constraint of Eq. (1) is defined: the optimization variable is the raw-magnitude displacement $\delta_i = m_i^{\mathrm{adv}} - m_i$, with $\delta_1 = 0$ and

$$|\delta_i| \;\le\; \epsilon_i \;=\; k\,e_i, \qquad k = 0.5, \tag{5}$$

the budget of Stage 1. Starting from $\boldsymbol{\delta}^{(0)} \sim \mathcal{U}(-\boldsymbol{\epsilon}, +\boldsymbol{\epsilon})$, the attack applies $S = 3$ iterations

$$\boldsymbol{\delta}^{(s+1)} \;\leftarrow\; \Pi_{\boldsymbol{\epsilon}}\Big(\boldsymbol{\delta}^{(s)} + \alpha_i \,\mathrm{sign}\,\nabla_{\boldsymbol{\delta}}\,\mathcal{L}_{\mathrm{CE}}(\pi_\phi(\boldsymbol{\mu}_\theta(\mathbf{x}'(\boldsymbol{\delta}^{(s)}))),\, y)\Big), \tag{6}$$

with per-point step $\alpha_i = 0.4\,\epsilon_i$, where $\mathbf{x}'(\boldsymbol{\delta})$ is the input whose magnitude channel is recomputed from the displaced magnitudes, $\Delta m_i^{\mathrm{adv}} = \Delta m_i + \delta_i - \delta_{i-1}$, and re-standardized with the clean statistics, and $\Pi_{\boldsymbol{\epsilon}}$ clips each δ_i to $[-\epsilon_i, +\epsilon_i]$. The cadence channel and $\sigma_{\Delta m}$ are never modified. Because the projection acts on the displacement at every iteration, Eq. (1) holds by construction throughout training (we also confirmed it empirically), and the adaptive attack of Sect. 5.7 uses the same parameterization, so every adversarial input in this paper satisfies it pointwise. The model is run in inference mode during the attack (no batch-normalization updates, no dropout), so that PGD does not pollute the training statistics; the resulting $\mathbf{x}_{\mathrm{adv}} = (\mathbf{x}_t, \mathbf{x}_m^{\mathrm{adv}})$ is then fed back in training mode, and this single adversarial counterpart is shared by all adversarial terms of the objective below.

3.2.4. Training objective: adversarial training plus representation-shaping regularizers

Let $\mathbf{z}$ and $\mathbf{z}_{\mathrm{adv}}$ be the encoder means for $\mathbf{x}$ and $\mathbf{x}_{\mathrm{adv}}$, $\mathbf{p}$ and $\mathbf{p}_{\mathrm{adv}}$ the classifier outputs, and $\mathbf{u}$ and $\mathbf{u}_{\mathrm{adv}}$ the (ℓ_2-normalized) projection-head embeddings. The training loss is

$$\mathcal{L} \;=\; \mathcal{L}_{\mathrm{CE}}^{\mathrm{cl}} \;+\; \lambda_{\mathrm{adv}}\,\mathcal{L}_{\mathrm{CE}}^{\mathrm{adv}} \;+\; \lambda_{\mathrm{tr}}\,\mathcal{L}_{\mathrm{TRADES}} \;+\; \lambda_{\mathrm{sc}}\,\mathcal{L}_{\mathrm{SupCon}} \;+\; \beta\,\mathcal{L}_{\mathrm{VIB}}, \tag{7}$$

with the following five terms. (i) The clean cross-entropy $\mathcal{L}_{\mathrm{CE}}^{\mathrm{cl}} = -\log p_y$ anchors clean accuracy. (ii) The adversarial cross-entropy $\mathcal{L}_{\mathrm{CE}}^{\mathrm{adv}} = -\log(p_{\mathrm{adv}})_y$ is the adversarial-training term of Madry et al. (2018). (iii) The TRADES consistency $\mathcal{L}_{\mathrm{TRADES}} = \mathrm{KL}(\mathbf{p}\,\|\,\mathbf{p}_{\mathrm{adv}})$ penalizes any prediction shift under perturbation (Zhang et al. 2019). (iv) The supervised adversarial contrastive loss (Khosla et al. 2020; Kim et al. 2020),

$$\mathcal{L}_{\mathrm{SupCon}} \;=\; -\sum_{i\in\mathcal{I}} \frac{1}{|P(i)|} \sum_{p\in P(i)} \log \frac{\exp(\mathbf{u}_i\cdot\mathbf{u}_p/\tau)}{\sum_{a\in A(i)}\exp(\mathbf{u}_i\cdot\mathbf{u}_a/\tau)}, \tag{8}$$

pulls clean and adversarial views of the same class together; $\mathcal{I}$ indexes the clean and adversarial views of the batch (two per light curve), $A(i) = \mathcal{I}\setminus\{i\}$, $P(i) \subseteq A(i)$ contains the views sharing the anchor's label (including its paired view), and $\tau = 0.1$. (v)

The VIB term $\mathcal{L}_{\mathrm{VIB}} = \mathrm{KL}(q_\theta(\mathbf{z} \mid \mathbf{x}) \,\|\, \mathcal{N}(\mathbf{0}, \mathbf{I}))$ compresses the latent representation (Alemi et al. 2017).

Terms (i)–(iii) constitute the defense proper: the minimax recipe of Madry et al. (2018), optionally smoothed by TRADES. Terms (iv) and (v) are representation-shaping regularizers motivated by the latent-space analysis of Appendix E.1; the ablation of Sect. 5.5 shows that they leave accuracy and robustness essentially unchanged, so the five-term configuration is a conservative default rather than five individually necessary defenses. The weights are fixed to $\lambda_{\mathrm{adv}} = 1.0$, $\lambda_{\mathrm{tr}} = 2.0$, $\lambda_{\mathrm{sc}} = 0.5$, and $\beta = 5 \times 10^{-4}$ for all three surveys; the projection head receives gradients only from term (iv).

3.2.5. Optimization protocol

The objective is minimized with Adam (Kingma & Ba 2015) (initial learning rate 5×10^{-4}, batch size 256) for up to 20 epochs per survey, halving the learning rate when the adversarial validation accuracy plateaus for four epochs and stopping early after eight (restoring the best weights). A separate model is trained per survey; no data are pooled.

3.3. Implementation and data availability

The pipeline is implemented in Python using TensorFlow 2 and Keras for the neural networks, `scikit-learn` (**?**) for the RF baseline, and `astropy.timeseries` (Astropy Collaboration et al. 2022) for the Lomb–Scargle periodogram. The period is not required by our classifier, which operates on raw differential sequences, nor by the RF, whose 16 features are period-independent (Appendix I); it is used only for catalog-level quality filtering, applied before any model comparison without class-dependent frequency windows, and for phase-folded visualizations. The adversarial light curves for the three surveys, the cached periods, the trained checkpoints, and the scripts reproducing every stage (surrogate training, adversarial generation, input construction, RRE training, and evaluation) are publicly available at `https://github.com/Gonzalo-Caceres/adversarial-light-curves`.

4. Datasets

We apply our method to three publicly available variable-star surveys with markedly different photometric properties: the optical *Optical Gravitational Lensing Experiment Survey III* the optical OGLE-III (Udalski 2004; Udalski et al. 2008) and *Gaia* DR2 (Gaia Collaboration et al. 2018) surveys, and the mid-infrared WISE survey (Wright et al. 2010). Together, these three surveys span complementary regimes of cadence, photometric uncertainty, and wavelength coverage, allowing us to assess the proposed defense under a wide range of observational conditions. For the three surveys we use the curated catalogs and light curves assembled by Becker et al. (2020), who compiled them for multi-survey variable-star classification; Table H.1 reports the native labels and counts of each catalog.

For all three surveys, we restrict the analysis to the four most populated variability classes available in each catalog and apply a per class random subsample (with a maximum of 20 000 light curves per class, seed 42) before the train–test split described in Section 5.1. The resulting class counts after the subsample are summarized in Table H.3.

4.1. Surveys

OGLE-III (Udalski 2004; Udalski et al. 2008) monitored the Galactic bulge, disk, and Magellanic Clouds in the I and V bands over multi-year baselines, producing well-sampled curves; labels follow the OGLE-III Catalog of Variable Stars (Soszyński et al. 2008, 2009a,b; Poleski et al. 2010; Graczyk et al. 2011). We use the four most populated classes – eclipsing binaries (ECL; aggregating EC, ESD, ED), RR Lyrae (RRLYR; RRab, RRc), Cepheids (CEP), and long-period variables (LPV; Miras, SRV, OSARG) – and adopt OGLE-III as the primary benchmark, as is standard in automatic variable-star classification (Aguirre et al. 2019; Becker et al. 2020; Donoso-Oliva et al. 2021; Bassi et al. 2021).

Gaia DR2: we use the all-sky classification of high-amplitude variable sources of *Gaia* Data Release 2 (Gaia Collaboration 2016; Gaia Collaboration et al. 2018; Holl et al. 2018; Rimoldini et al. 2019), with G-band light curves. *Gaia* curves have few observations per source (typically ∼20–30) and small per-point photometric errors, which restricts the admissible perturbation budget and provides a complementary regime for the defense. The working classes are RRLYR (RRAB, RRC, RRD, ARRD), LPV (MIRA_SR), CEP (classical, Type II, anomalous), and δ Scuti / SX Phoenicis (DSCT). We adopt DR2 rather than DR3 – which expands the variability products to 10.5 million sources with epoch photometry and 12.4 million classified (Eyer et al. 2023; Rimoldini et al. 2023) – for comparability with the compilation of Becker et al. (2020), on which our preprocessing, labels, and baselines are built; since DR3 uncertainties are typically smaller, DR2 is also the more conservative stress test.

WISE (Wright et al. 2010): we use $W1$-band ($3.4\,\mu$m) light curves with labels inherited from the OGLE-III nomenclature (Becker et al. 2020), grouped into the same four classes as *Gaia*. WISE is the noisiest of the three surveys, so the photometric budget – and hence the attack surface – is largest there; its CEP and DSCT classes are heavily under-represented in the raw catalog (Table H.3), which manifests itself in the per-class results. The native class labels and counts of the three catalogs are reported in Appendix H.

4.2. Summary of class counts after subsampling

Table H.3 summarizes the per-class counts entering our experiments. WISE is the most skewed (CEP and DSCT contribute fewer than 3,000 curves *combined*), *Gaia* less so, and OGLE is the most balanced of the three. The final per-class train/test counts after the split, together with the per-object epoch distribution of each survey, are listed in Table H.2 (Appendix H).

5. Results

5.1. Experimental setup and evaluation protocol

For each survey, the per-class subsample of Sect. 4 is split into training, validation, and test subsets (70%–10%–20%, stratified by class, seed 42). The validation subset is used only for early stopping and learning-rate scheduling (Sect. 3.2.5); the test subset is never seen during training or model selection. All classifiers share the same split.

The main cross-survey comparison uses three classifiers: an RF on hand-engineered features (FATS; Nun et al. 2015), the two-tower CNN of Sect. 3.1.1, and the proposed RRE of Sect. 3.2. A GRU recurrent baseline follow-

ing Donoso-Oliva et al. (2021) is reported only in the model-specific OGLE-III attack sweep. The RF and CNN are trained on the clean training set with the standard cross-entropy; the RRE is trained with the full objective of Eq. (7), generating a fresh adversarial counterpart of every training batch on the fly (Sect. 3.2.3).

Robustness is assessed on adversarially perturbed versions of each test set, generated as in Sect. 3.1.2 with $k = 0.5$, and all classifiers are evaluated on the same pair of clean and adversarial test sets; we verified on every generated set that the perturbations satisfy Eq. (1) exactly (Appendix A). For each model and survey we report the clean accuracy $\mathrm{acc_c}$, the adversarial accuracy $\mathrm{acc_a}$, and the drop $\mathrm{acc_a} - \mathrm{acc_c}$ in percentage points (pp). Every metric carries a 95% percentile-bootstrap confidence interval over the test set ($B = 2{,}000$); the clean→adversarial drop and the RRE−CNN gap are recomputed within each replicate (a *paired* bootstrap), accounting for the correlation between the two estimators. Alongside accuracy we report balanced accuracy and macro-averaged F_1, which weight every class equally and expose minority-class behavior that accuracy can mask (Tables C.1 and C.2).

We first quantify the vulnerability of the baselines under attacks tailored to each model (Sect. 5.2), then evaluate RRE across the three surveys (Sect. 5.3), ablate its loss components (Sect. 5.5), and test it against a fully adaptive white-box attack as a first-order check against gradient masking (Sect. 5.7; Athalye et al. 2018; Carlini et al. 2019). Per-class breakdowns and latent-space evidence are given in Appendices B and E.1.

5.2. Effectiveness of the adversarial attack on supervised baselines

To motivate the defense presented in the next subsection, we first show that the photometrically bounded perturbations of Section 3.1.2 are an effective stress test for the standard light-curve classifiers used as baselines. The astronomical constraint is the same throughout: only the observed magnitudes are displaced, and every proposed light curve must satisfy the pointwise photometric bound $|m_i^{\mathrm{adv}} - m_i| \leq k e_i$. The numerical search used to find such a curve depends on the classifier only for practical reasons. For the CNN and RNN we use a gradient-based PGD search; for the Random Forest, whose features are recomputed from the perturbed photometry, we use the standard query-based Square Attack (Andriushchenko et al. 2020). Thus all baselines are tested under the same photometric threat model, even though the optimizer used to find the bounded perturbation differs. We report the default budget $k = 0.5$ adopted throughout the paper and, additionally, the full one-reported-error budget $k = 1.0$, corresponding to a displacement of one full reported uncertainty per point.

5.2.1. Baseline sensitivity under the same photometric bound

Under this gradient-based search, the CNN accuracy decreases from 0.964 to 0.658 on OGLE-III (−30.6 pp) and from 0.914 to 0.821 on WISE (−9.3 pp), while on *Gaia* DR2 it only drops from 0.868 to 0.850 (−1.7 pp): the small *Gaia* uncertainties shrink the admissible budget $k\,e_i$ itself (Table 1). On OGLE-III and WISE the perturbations push several individual classes below random-guess accuracy (Appendix B).

The same gradient-based search is applied to the RNN on OGLE-III, where the trained instance used for this sweep reaches a clean accuracy of 0.950. Under the PGD attack at $k = 0.5$, its accuracy declines to 0.592 (−35.8 pp), indicating that the recurrent baseline is also vulnerable to photometrically bounded perturbations.

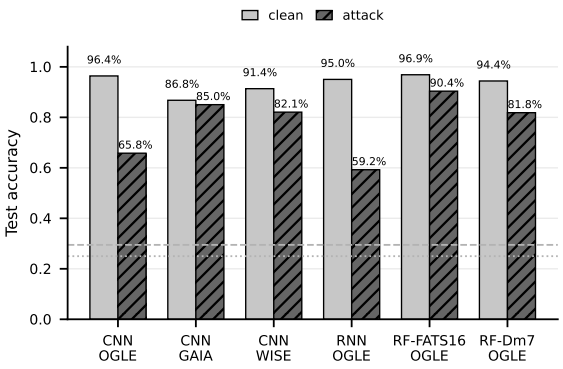


Fig. 2: Classification accuracy of the supervised baselines under photometrically bounded adversarial searches with per-point budget $k = 0.5$. The CNN is evaluated on all three surveys; the RNN and two RF variants are shown on OGLE-III. The numerical search is gradient-based for the CNN/RNN and query-based for the RFs, but every perturbed light curve obeys the same pointwise photometric bound. Numbers above the bars give the corresponding test accuracies in percent. The dotted line marks the random-guess level for a 4-class problem; the dashed line marks the majority-class rate of the OGLE-III test set (0.29), the more informative floor for an imbalanced split. The transferred CNN-targeted cross-survey comparison is reported separately in Table 1.

For the Random Forest we apply the same photometric logic with a query-based search, since its hand-engineered feature map is not differentiated. We use Square Attack (Andriushchenko et al. 2020) on OGLE-III, against two variants. For the actual 16-feature FATS RF of Table 1 (RF-FATS16), every query perturbs the raw magnitudes of the full light curve within the pointwise budget and recomputes the complete feature vector, so the attack targets exactly the classifier reported in the main results. On a stratified subsample of 800 test curves (clean accuracy 0.969), the attack reduces the accuracy to 0.904 (95% bootstrap CI $[0.882, 0.924]$; −6.5 pp) at $k = 0.5$ and to 0.805 ($[0.779, 0.833]$; −16.4 pp) at $k = 1.0$, plateauing after ∼550 queries. We use a subsample because each query recomputes the full feature vector, which makes the attack computationally expensive; the bootstrap intervals confirm that the accuracy drops are well resolved despite the reduced sample size. A simplified 7-feature variant in standardized Δm space (RF-$\Delta m7$) falls from 0.944 to 0.818 (−12.6 pp) at $k = 0.5$. Hand-engineered aggregate statistics therefore confer real but partial protection: the RF degrades two to five times less than the deep baselines, yet its apparent immunity under the transferred CNN-targeted perturbations of Table 1 does not survive an attack that targets its own feature space within the same photometric threat model.

Sensitivity to the perturbation budget. At the maximal budget considered, $k = 1.0$ (one full reported uncertainty per point) every baseline degrades monotonically: the CNN falls to 0.465 (−49.9 pp), the RNN to 0.427 (−52.4 pp), RF-FATS16 to 0.805 (−16.4 pp), and RF-$\Delta m7$ to 0.724 (−22.0 pp); the two neural baselines approach the four-class random-guess level. Vulnerability to photometrically bounded perturbations is therefore not a

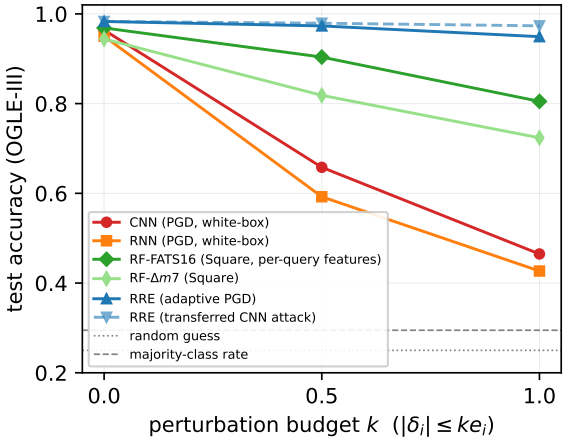


Fig. 3: Accuracy on OGLE-III as a function of the photometric perturbation budget k under the same pointwise magnitude bound (solid lines): gradient-based searches for the CNN and RNN, query-based searches for the RF variants (RF-FATS16 with per-query feature recomputation), and adaptive PGD for RRE (at $k = 0.5$ and $k = 1.0$). The dashed line shows RRE under the *transferred* CNN-surrogate attack across the full admissible range; horizontal reference lines mark the four-class random-guess level and the majority-class rate of the test set. Adaptive white-box evaluation of RRE on all three surveys is reported separately in Sect. 5.7.

property of any single architecture — it affects every supervised classifier evaluated here, to varying degrees, under gradient-based and query-based attacks. Table D.1 in Appendix D collects these baseline stress tests at both budgets, and Fig. 3 summarizes them.

A probe of a pre-trained transformer. The baselines above are the CNN, GRU, and feature-based classifiers that have dominated variable-star classification; the current literature is increasingly moving towards self-supervised transformer representations (Section 2). As an exploratory probe — not as a performance-matched baseline — we therefore also attacked ASTROMER 2 (Donoso-Oliva et al. 2026), a transformer foundation model pre-trained on millions of light curves that, notably, weights its pre-training loss by the reported photometric uncertainties, making it *a priori* the most plausible candidate for intrinsic robustness to a photometric attack. We fine-tuned it end-to-end on the OGLE-III training split following its published two-phase protocol (frozen-encoder head training, then full fine-tuning at a low learning rate). It reaches a clean accuracy of 0.776 on OGLE-III — below the from-scratch CNN and RF, which is expected rather than anomalous: foundation models of this kind are designed for the few-shot regime (Donoso-Oliva et al. 2026), whereas our setup is label-rich (up to 20,000 curves per class), the regime in which task-specific supervised models are typically strongest. We therefore do not treat ASTROMER 2 as a state-of-the-art performer on this task, but as a representative modern architecture. Under the transferred CNN-targeted attack its accuracy falls to 0.249 at the maximal budget $k = 1.0$ (drop -52.6 pp, 95% bootstrap CI $[-54.0, -51.2]$), comparable to the collapse of the CNN and RNN in the same setting (Table D.1). Even an uncertainty-weighted transformer representation, then, does not confer robustness within the photometric budget: the vulnerability documented here is not an artifact of older architectures.

5.3. Robustness of the proposed RRE classifier across surveys

We now evaluate the Robust Raw Encoder (RRE) of Section 3.2 on the same clean and adversarial test sets used for the baseline classifiers in Section 5.2. The model operates directly on the two-tower $(\Delta t, \Delta m, \sigma)$ representation of the raw light curve. At inference time, the RRE receives either the clean or the adversarial counterpart of a given light curve and outputs class probabilities through its deterministic encoder mean and classifier head (Section 3.2.2).

Table 1 compares the clean and adversarial accuracy of the CNN surrogate, the feature-based RF baseline, and RRE on the three surveys. The recurrent baseline is kept as an auxiliary OGLE-only model-specific check in Table D.1, rather than as a cross-survey comparator. The RRE preserves its clean accuracy under the CNN-targeted attack on all three: the drop is -0.4 pp on OGLE-III ($0.983 \to 0.979$), -0.5 pp on *Gaia* DR2 ($0.911 \to 0.906$), and -0.4 pp on WISE ($0.932 \to 0.928$).

A paired bootstrap (Appendix C) attaches 95% confidence intervals to these numbers: the clean→adversarial drop of RRE is +0.4, +0.5, and +0.4 pp on OGLE-III, *Gaia*, and WISE, respectively—each interval excludes zero yet is a small fraction of the corresponding CNN drop—and the adversarial-accuracy advantage of RRE over the CNN (+32.1, +5.6, and +10.7 pp) is significant on all three surveys. The same stability appears in class-balanced metrics: RRE balanced accuracy changes from 97.9 to 97.5 on OGLE-III, 88.5 to 88.0 on *Gaia* DR2, and 61.0 to 60.3 on WISE, while macro-F_1 changes from 98.1 to 97.7, 89.3 to 88.7, and 62.3 to 61.5, respectively (Table C.1).

Survey-by-survey behavior. On OGLE-III — where the attack is most damaging — the RRE keeps 0.979 under attack, essentially closing the 30 pp robustness gap of the CNN. On *Gaia* DR2 the small per-point uncertainties shrink the budget for every model (the CNN drops only -1.7 pp). On WISE, the noisiest survey and largest attack surface, the RRE maintains $0.932 \to 0.928$; its aggregate accuracy is driven by the well-represented RRLYR and LPV classes (Section 5.4), which is why balanced accuracy and macro-F_1 are reported throughout.

Reading the clean-accuracy column. The clean-accuracy advantage of RRE over the CNN (+1.9 pp on OGLE-III) should not be attributed to the defense: the CNN lacks the $\sigma_{\Delta m}$ channel, so the models differ in inputs and training. The architecture-matched clean control is V1, which reaches 0.988 on OGLE-III — slightly *above* the defended 0.983, consistent with the small robustness–accuracy trade-off of Sect. 5.5. The informative content of the table is the stability under attack, not the clean ranking.

Does the representation matter, or only the training? The AT-CNN row answers this directly: applying the exact adversarial-training recipe of the V2 variant to the surrogate CNN architecture recovers most of its robustness (on OGLE-III, from 0.658 under its own white-box attack to 0.904 under an equally aggressive adaptive attack, $T = 40$, five restarts), but it does not close the gap to the raw-representation models: across the three surveys the AT-CNN remains 3–7 pp below V2 and RRE in both clean and adaptive-attack accuracy (e.g. OGLE-III 0.954 vs. 0.983 clean, 0.904 vs. 0.973 adaptive), and it pays the classical clean-accuracy cost of adversarial training (-1.0 pp on

Table 1: Clean and adversarial classification accuracy across the three surveys under the transferred CNN-targeted attack of Section 5.2 with $k = 0.5$. Each block of three columns corresponds to one survey. Two controls isolate the sources of the RRE robustness: AT-CNN is the *same* surrogate CNN retrained with the adversarial-training recipe of V2 (isolating the training), and V1 is the RRE architecture trained with clean cross-entropy only (isolating the representation). Model-specific adaptive attacks are reported in Sect. 5.5. The RF entries are transfer-attack results, not robustness certificates: under a model-specific black-box Square Attack the same OGLE-III RF drops to 0.904 at $k = 0.5$ (Table D.1). Drop = $\mathrm{acc_a} - \mathrm{acc_c}$ in pp.

	OGLE-III			*Gaia* DR2			WISE		
Model	$\mathrm{acc_c}$	$\mathrm{acc_a}$	drop	$\mathrm{acc_c}$	$\mathrm{acc_a}$	drop	$\mathrm{acc_c}$	$\mathrm{acc_a}$	drop
CNN	0.964	0.658	−30.6	0.868	0.850	−1.7	0.914	0.821	−9.3
AT-CNN	0.954	0.933	−2.1	0.871	0.857	−1.4	0.916	0.903	−1.2
RF	0.976	0.925	−5.0	0.923	0.921	−0.2	0.926	0.903	−2.3
V1 (clean control)	0.988	0.916	−7.2	0.919	0.914	−0.5	0.934	0.896	−3.8
RRE (ours)	0.983	0.979	−0.4	0.911	0.906	−0.5	0.932	0.928	−0.4

OGLE-III) that the raw encoder largely avoids. Standard adversarial training is therefore the operative defense, but the raw representation contributes accuracy and robustness that training alone does not supply — with the caveat that AT-CNN also differs in capacity and lacks the $\sigma_{\Delta m}$ channel, so this is evidence for the representation rather than a controlled attribution. The 3–7 pp gaps are, moreover, well outside the training-seed scatter of the defended variants, which is 0.07 pp across five seeds on OGLE-III (Section 5.5), so single-run comparisons are adequate.

5.4. Per-class behavior

Appendix B reports per-class accuracies and F_1 scores with bootstrap intervals. Robustness is uniform: no class on any survey loses more than ∼2 pp of accuracy or F_1 under the transferred attack, so the aggregate stability of Table 1 is not an average over heterogeneous behavior. The known limitation is WISE, whose under-represented CEP and DSCT classes (Table H.1) are poorly learned or not learned at all on *clean* data by every classifier evaluated here, a clean-trainability effect of catalog imbalance rather than an adversarial vulnerability. We therefore claim no useful absolute performance for WISE CEP and DSCT, for which robustness is vacuous.

5.5. Ablation study of the V5 loss components

To quantify the contribution of each loss component of Eq. (7), we trained five variants of RRE that add the terms one at a time: V1 uses clean cross-entropy only (a standard supervised classifier on the raw two-tower representation); V2 adds adversarial cross-entropy (Madry-style adversarial training; Madry et al. 2018); V3 adds TRADES; V4 adds the supervised adversarial-contrastive term; and V5 adds the VIB, completing the full model (the RRE of Table 1). Every variant is trained from scratch on the OGLE-III training split with the same encoder, optimization protocol (Sect. 3.2.5), and inner attack (Sect. 3.2.3); only the loss changes. We evaluate each on the clean test set, on the same transferred CNN-targeted adversarial set used throughout the paper, and under the adaptive white-box attack of Sect. 5.7 computed on its own gradient (Table 2).

The V1 variant, which contains no adversarial component, reaches 0.988 clean accuracy but loses 7.2 pp under the transferred attack. This is already far less than the 30.6 pp lost by the CNN baseline (Sect. 5.2), because the two-tower architecture and the per-point budget restrict how far the magnitude tower can be pushed; but the advantage is transfer-specific: under its own adaptive attack V1 collapses to 0.699 (−28.9 pp), comparable to the CNN under its own white-box attack (0.658). The raw representation on its own is therefore not robust and, lacking the smoothing that phase binning would provide, must obtain all of its robustness from the loss.

Table 2: Ablation of the loss components, trained and evaluated directly on the raw two-tower $(\Delta t, \Delta m, \sigma)$ representation. Top block: the incremental OGLE-III ladder (each row adds one term to the previous). Bottom blocks: the V1/V2/V5 comparison repeated on *Gaia* DR2 and WISE with identical protocols. Transf. = transferred CNN-targeted attack; Adapt. = adaptive white-box PGD on each variant's own gradient (OGLE-III: $T = 20$, 3 restarts; *Gaia*/WISE: $T = 40$, 5 restarts; $k = 0.5$). Single runs (seed 42); five-seed means for V1, V2 and V5 on OGLE-III are in the text (V1: 0.9000 ± 0.0122).

Survey	Variant	$\mathrm{acc_c}$	Transf.	Adapt.
OGLE-III	V1 (CE only)	0.988	0.916	0.699
	V2 (+ AdvCE)	0.984	0.979	0.971
	V3 (+ TRADES)	0.982	0.979	0.973
	V4 (+ SupCon)	0.984	0.982	0.976
	V5 (+ VIB; full)	0.983	0.979	0.973
Gaia DR2	V1	0.919	0.914	0.912
	V2	0.922	0.918	0.916
	V5	0.911	0.906	0.904
WISE	V1	0.934	0.896	0.840
	V2	0.935	0.930	0.926
	V5	0.932	0.928	0.925

Adding adversarial cross-entropy (V2) is the dominant step: it lifts the transferred adversarial accuracy from 0.916 to 0.979 (−7.2 → −0.5 pp) and the adaptive one from 0.699 to 0.971, at a clean-accuracy cost of ∼0.4 pp (0.988 → 0.984). The accuracy–robustness trade-off of adversarial training (Zhang et al. 2019) is thus present but small, because the photometric budget is small relative to the intrinsic variability the encoder must model.

The three remaining components leave both accuracies essentially unchanged: from V2 to V5 the clean accuracy stays within 0.2 pp of 0.984 and the transferred drop between −0.3 and −0.5 pp; under the adaptive attack the latent regularizers add a small margin (0.971 → 0.976 for V4), partly given back by the VIB (0.973 for V5). These V2–V5 differences of a few tenths of a point should not be over-interpreted; all four sit far above the 0.699 of V1.

The conclusion holds across surveys. Because a 0.2-pp OGLE-III gap could hide a regime where the additional terms matter, we repeated the V1/V2/V5 comparison on the other two

surveys (Table 2, bottom blocks). V2 matches or exceeds V5 on every survey and attack (on *Gaia* it is ∼1 pp *better*, an order of magnitude above the OGLE-III seed scatter), so no survey provides a regime where the additional terms improve on plain adversarial training. The table also shows that vulnerability scales with the error budget: under its own adaptive attack the undefended V1 loses 28.9 pp on OGLE-III and 9.4 pp on WISE, but only 0.8 pp on *Gaia*, whose tiny uncertainties leave almost no admissible perturbation — there, the instrument, not the training, is the defense. This is not an artifact of ignoring the faint end, where *Gaia* uncertainties grow (e.g., Fig. 14 of Rimoldini et al. 2019): in a random sample of 4,000 of our curves the per-curve median G-band uncertainty rises from ∼0.001 mag at $G < 15$ to ∼0.010 mag at $G \in [19, 21)$ and 0.015–0.027 mag for the faintest sources ($G \approx 20.7$), still comparable to the 90th percentile of the OGLE-III budgets (0.033 mag) and about half the WISE median (0.041 mag), while two thirds of the *Gaia* curves ($G < 18$) sit an order of magnitude below the OGLE-III median (0.006 mag).

To quantify the training-seed sensitivity, we repeated the V1, V2, and V5 runs with five independent seeds: the seed-to-seed scatter of the adversarial accuracy is 0.07 pp for both V2 (0.9785 ± 0.0007) and V5 (0.9790 ± 0.0005), against 1.2 pp for V1 (0.9000 ± 0.0122), so adversarial training also stabilizes robustness across runs. Complementary controls that activate each loss family *without* adversarial cross-entropy, and a VIB-only β-sweep, are collected in Appendix F: TRADES alone matches V2 under the transferred attack, SupCon alone is only partially robust, and VIB alone is *not* robust (−8.0 to −9.4 pp, worsening with compression strength), corroborating Korshunova et al. (2021); random-noise augmentation is tested in Sect. 5.6. We therefore retain all five components as a conservative full configuration, not as five individually necessary defenses: AdvCE and TRADES supply the predictive clean–adversarial coupling, while SupCon and VIB regularize the latent geometry, which Appendix E.1 shows to be stable under perturbation, not only at the classifier output.

5.6. Is random noise augmentation enough?

A perturbation bounded by the photometric error is, by construction, pointwise compatible with the reported uncertainties, so the most natural alternative to adversarial training is to augment the training set with *random* displacements drawn inside the same budget, $\delta_i \sim \mathcal{U}(-k\,e_i, +k\,e_i)$, or a Gaussian of matching scale clipped to the same budget, applied with the same raw-magnitude parameterization as the attack and resampled at every batch. We trained the V1 encoder (clean cross-entropy only) with each augmentation and evaluated it under the identical transferred-attack protocol. Random augmentation provides no measurable protection: the uniform-noise model reaches 0.987 clean but 0.914 adversarial accuracy (−7.3 pp), and the Gaussian variant 0.987/0.923 (−6.4 pp) — statistically indistinguishable from the un-augmented V1 baseline (0.9873 ± 0.0007 clean, 0.9000 ± 0.0122 adversarial across five seeds; Section 5.5), and an order of magnitude worse than adversarial training (−0.5 pp for V2). Exploring the photometric budget on *average* is therefore not a substitute for optimizing against its *worst case*: the robustness of RRE is driven by adversarial training, not by mere exposure to budget-scale input variability.

5.7. Adaptive white-box attack on the RRE classifier

Robustness under a transferred attack can be illusory: defenses must be evaluated under *adaptive* attacks whose gradient is computed on the defended classifier itself (Carlini et al. 2019). We therefore attack the RRE with a white-box PGD back-propagated through its own encoder and classifier head, using the same raw-magnitude displacement parameterization as the inner training attack (Section 3.2.3): $\delta_i = m_i^{\mathrm{adv}} - m_i$ clipped to $|\delta_i| \le k\,e_i$ ($k = 0.5$) at every iteration, with Δm recomputed by differencing and Δt and the uncertainty channel fixed, so Eq. (1) holds by construction. The attack maximizes the true-label cross-entropy far more aggressively than the training attack: $T = 40$ iterations, per-point step $\alpha = 0.1\,\epsilon_i$, and 5 random restarts (keeping the worst-case restart per curve), versus $S = 3$ single-restart iterations during training.

Under the adaptive PGD attack, the RRE drops from a clean accuracy of 0.983 to an adversarial accuracy of 0.973 on the OGLE-III test set, a drop of −1.0 pp, against −0.4 pp under the transferred attack. The per-class drops range from −0.1 pp on LPV (the largest and most well-sampled class, 0.998 → 0.998) to −1.8 pp on ECL (0.977 → 0.959), with RRLYR at −1.0 pp and CEP at −1.3 pp. Even this largest per-class drop is more than an order of magnitude smaller than the per-class collapses suffered by the CNN baseline under its own white-box PGD attack (Section 5.2).

This conclusion is not specific to OGLE-III. Table G.1 repeats the adaptive white-box PGD evaluation against RRE's own gradient on all three surveys; the accuracy drop under the adaptive attack does not exceed one percentage point anywhere—−1.0 pp on OGLE-III, −0.7 pp on *Gaia* and −0.7 pp on WISE—so the robustness survives a first-order attack that directly follows the defended model's gradient. This does not certify robustness, but it makes a simple gradient-masking or transfer-artifact explanation unlikely (Athalye et al. 2018; Carlini et al. 2019). At the maximal budget considered ($k = 1.0$) the adaptive attack reduces the OGLE-III accuracy to 0.949 (−3.4 pp; Fig. 3).

The adaptive attack is roughly 2.5 times stronger than the transferred one (−1.0 versus −0.4 pp), yet its drop remains more than thirty times smaller than the CNN's white-box collapse: the robustness of Section 5.3 is structural, not an artifact of gradient transferability. We do not claim certified robustness — empirical first-order robustness is the strongest statement an adaptive PGD evaluation supports.

5.8. Gradient-free, margin-based, and attack-potency checks

Following the checklist of Carlini et al. (2019), three further checks (protocols in Appendix G) complete the evaluation. A gradient-free Square-style attack on the RRE probabilities (800 curves, 1,000 iterations, $k = 0.5$) costs only −0.25 pp, versus −0.37 pp for the adaptive white-box PGD run on the same subsample: the gradient-based attack remains the stronger one, the signature of informative, unmasked gradients despite the stochastic bottleneck (Korshunova et al. 2021; Athalye et al. 2018). The margin-based difference of logits ratio (DLR) loss of Auto-PGD (Croce & Hein 2020) yields 0.973 at $T = 40$ and $T = 100$, identical to cross-entropy: the attack is converged and the margin objective finds nothing new. Finally, inflating the budget to $k = 5$ and $k = 20$ — far outside the threat model — collapses the accuracy to 0.135 and 0.007: the attack annihilates the model when given room, confirming that robustness at admissible budgets reflects the threat model, not an weak attack.

6. Discussion

The adversarial perturbations considered here are pointwise consistent with the reported uncertainties: every perturbed magnitude stays within $k\,e_i$ of its observed value, with a fixed budget $k = 0.5$ by default and at most $k = 1$, so that no measurement is displaced by more than its own reported 1σ error. The threat model is therefore bounded by construction. This bound defines a deterministic uncertainty set rather than a physical hard limit — a 1σ error is the scale of a distribution, not an interval, and genuine noise exceeds it for roughly a third of the measurements — so the perturbations remain well inside the range of displacements that the reported uncertainties already allow. They are not a stochastic draw from the survey noise model, but a controlled worst-case stress test: would the classifier assign the same label to every light curve whose measurements are pointwise consistent with the reported error bars? If not, the prediction is sensitive to photometric uncertainty regardless of its clean-test confidence. Pointwise admissibility is deliberately permissive: the optimized perturbations are temporally correlated and hence statistically distinguishable from independent noise (Appendix A).

This makes the framework a model-auditing tool: photometrically bounded adversarial accuracy complements clean accuracy when validating classifiers, and the clean-to-adversarial prediction change is a per-source *robustness flag* (Table E.1; Fig. F.1). Under the transferred attack, 30.7%, 1.8%, and 9.8% of CNN test predictions change on OGLE-III, *Gaia*, and WISE, against 0.4–0.5% for RRE. For the defended model the flagged sources are the low-margin ones (median clean margin 0.06–0.18, versus 0.93–1.00 for stable sources), so the flag routes ∼0.5% of sources to follow-up. The undefended CNN shows the opposite, more dangerous pattern: on OGLE-III its flipped sources have a median clean margin of 0.98, so a confidence cut cannot substitute for the adversarial audit. These flip rates use the true-label attack of the paper; in deployment the same flag is obtained by attacking the clean *predicted* class, or by minimizing the margin $p_1 - p_2$, a label-free variant whose systematic evaluation we leave to future work.

Flagged sources are the noisiest ones. The CNN-flagged OGLE-III sources have a median reported error of 0.053 mag against 0.007 mag for the stable ones (a factor of ∼8; Mann–Whitney $p \ll 10^{-100}$; Fig. E.3) and a lower per-curve signal-to-noise ratio (60 vs. 86). The association is partly mechanical — the budget is $k\,e_i$, so noisier sources have more room to be pushed across the boundary, and the S/N proxy contains e_i — so we do not claim it as an independent validation of the flag. It does show that the flag fires on the noisiest objects rather than on well-measured ones, the desirable behavior for a follow-up trigger. Testing it against quality indicators *independent* of e_i (blending or crowding flags, light-curve-fit χ^2, disagreement with independent classifications) is a natural next step.

Scope of the threat model. The adversarial curves are generated by a surrogate CNN and transferred, so robustness against the surrogate need not imply robustness against an attacker with gradient access to RRE itself; we therefore complement the transferred evaluation with the adaptive white-box attack of Sect. 5.7. The perturbation set is a per-point box, $|m_i^{\mathrm{adv}} - m_i| \leq k\,e_i$, aligned with the reported uncertainties; it does not cover structured, ℓ_2-budgeted, or time-correlated perturbations, nor perturbations to the sampling times t_i. The robustness we report is also conditional on well-calibrated uncertainties: under- or over-estimated error bars would shift the admissible set accordingly. The k-sweep of Fig. 3 quantifies this: attacking the $k = 0.5$-trained model at $k = 1.0$ — equivalent to uncertainties underestimated twofold — lowers the adaptive OGLE-III accuracy only from 0.973 to 0.949. The caveat is most relevant for WISE: in crowded fields its $W1$ photometry is affected by source confusion and blending, so the reported per-epoch uncertainties, which do not fully capture confusion noise, can underestimate the true scatter (**??**). Our choice of *Gaia* DR2 over DR3 (Sect. 4) is conservative in this respect: the smaller DR3 uncertainties would shrink the admissible set further, reinforcing the finding that *Gaia*-like photometry is nearly immune to the attack. We regard these as boundaries of the present claims rather than as evidence of unconditional robustness.

Outlook for upcoming surveys. The setting becomes operationally relevant with LSST (Ivezić et al. 2019), whose alert brokers must classify millions of variable sources per night from sparse, heteroskedastic photometry. Worst-case-within-errors evaluation offers such pipelines a training-time defense costing ∼0.5 pp of clean accuracy and an offline robustness audit identifying the ∼0.5% of labels that are not stable within their own error bars. Because the attack surface scales with the photometric errors, the audit matters most at the faint end, where the science is new and the uncertainties are largest.

7. Conclusions

We studied photometric uncertainty as an attack surface for machine-learned variable-star classifiers. Adversarial light curves that displace each magnitude only within its reported error degrade every standard classifier we evaluated: on OGLE-III the CNN loses 30.6 percentage points (pp), the GRU 35.8 pp, and the non-differentiable 16-feature RF 6.5 pp at $k = 0.5$ under a query-only black-box Square Attack; an exploratory probe of a fine-tuned ASTROMER 2 transformer is also affected. The vulnerability is therefore not an artifact of a particular architecture, although aggregate feature representations attenuate it substantially relative to deep sequence models.

As a defense we proposed the Robust Raw Encoder (RRE), a two-tower convolutional classifier on the raw differential representation $(\Delta t, \Delta m, \sigma)$, trained adversarially with the per-point photometric error as the budget of an inner projected-gradient attack. On OGLE-III, *Gaia* DR2, and WISE it preserves its clean accuracy under attack (drops of 0.4–0.5 pp, statistically detectable but operationally negligible; Appendix C), and under an adaptive white-box attack on its own gradient (Carlini et al. 2019) it loses only 1.0 pp on OGLE-III, an order of magnitude less than the CNN under its own attack. The ablation, repeated on all three surveys, shows that predictive coupling between clean and adversarial views is the operative ingredient: plain adversarial training (V2) matches or exceeds the full objective everywhere at a clean-accuracy cost of ∼0.4 pp, the remaining terms act as latent-space regularizers, and VIB alone provides no robustness (corroborating Korshunova et al. 2021). Random-noise augmentation within the same budget gives no protection either, so robustness requires optimizing against the worst case within the error bars. The representation also matters: an adversarially trained CNN stays 3–7 pp below the raw-representation models on every survey. Vulnerability scales with the error budget — on *Gaia* even the undefended encoder loses less than one point — and the defended model passes the full evaluation checklist (Sect. 5.8), with clean and adversarial views

mapped onto the same latent region. Its main limitation is not adversarial: the under-represented WISE classes are not learned by any classifier, a catalog property (Appendix B).

Future work will extend this training to further architectures, surveys, and threat models. A natural next step is multiband classification (e.g. Cabrera-Vives et al. 2024), where the box constraint becomes per band, $|m_{i,b}^{\rm adv} - m_{i,b}| \leq k\, e_{i,b}$, and a new question arises: whether perturbations must preserve cross-band (colour) coherence, a physically motivated restriction with no single-band analogue. The audit itself is also amenable to automation: an agentic pipeline could orchestrate gradient- and query-based attacks across budgets and survey releases, while language models could propose structured, physically motivated perturbation families (zero-point drifts, seeing- or blending-dependent systematics) beyond the per-point box, to be then optimized by the bounded search presented here.

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Data availability

The full analysis pipeline and the scripts that reproduce every experiment are publicly available at `https://github.com/Gonzalo-Caceres/adversarial-light-curves`. The derived data products — the generated adversarial light curves for the three surveys, the cached Lomb–Scargle periods used by the Random Forest baseline, and the trained model checkpoints — are archived in a Zenodo deposit (DOI: `10.5281/zenodo.XXXXXXX`). The underlying light curves are available from the public OGLE-III, *Gaia* DR2, and WISE archives, as compiled by Becker et al. (2020).

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Appendix A: Structure, detectability, and budget validation of the perturbations

We measured the temporal structure of the perturbations performed $\boldsymbol{\delta} = \mathbf{m}^{\rm adv} - \mathbf{m}$ directly on the attacked test sets. At the full perturbation fraction the lag-one autocorrelation of $\boldsymbol{\delta}$ has a median of 0.77–0.86 across surrogates (a null within-curve permutation gives −0.005), 96–99% of adjacent perturbed points share the same sign (white noise: 50%), and the normalized curvature $\sum_i(\Delta^2\delta_i)^2 / \sum_i \delta_i^2$ is suppressed by a factor of 5–10 relative to the shuffled null. Removing the smoothness term does not remove the structure entirely; for the *Gaia* attack ($\lambda_{\rm sm} = 0$) the median autocorrelation lag-one is still 0.26 — indicating that the classifier gradients are themselves temporally correlated, with $\lambda_{\rm sm}$ amplifying the effect. At the same time, the perturbations exhaust the photometric budget pointwise: the median of $|\delta_i|/(ke_i)$ is 1.0, with 93–98% of perturbed points saturating the bound.

This combination — pointwise admissible amplitudes with non-noise-like temporal ordering — determines how the attack can and cannot be detected (Table A.1). The distributions of scale-sensitive features in the test set (standard deviation, robust amplitude, Stetson K) are statistically indistinguishable between clean and attacked curves (two-sample KS $D \leq 0.022$, $p > 0.08$), so that a survey pipeline monitoring amplitude-like statistics would not notice the attack. Order-sensitive statistics, in contrast, expose it decisively: the attack shifts the von Neumann ratio $\eta_{\rm vN}$ (median 1.81 → 1.64) and the lag-one autocorrelation of the magnitude sequence (median 0.09 → 0.17), yielding KS statistics of 0.07–0.10 with p-values between 10^{-13} and 10^{-25} at full perturbation fraction, consistently across the OGLE attacks shown below, with a monotonic dose–response as the perturbation fraction grows. For the *Gaia* configuration ($k = 0.5$) no feature is significantly displaced, showing that detectability also scales with the budget. These measurements sharpen the sense in which the attack is pointwise admissible — each magnitude stays within its error bar — while motivating a cheap audit: flagging curves whose $\eta_{\rm vN}$ or autocorrelation is anomalous for their class provides a first-line *robustness flag* against smooth bounded attacks, which we return to in the Discussion.

A.1. Budget validation on the generated test sets

Before evaluating classifier robustness, we verify empirically that the adversarial perturbations satisfy the photometric-error budget of Eq. (1). For every observation i in the adversarial test set of a given survey, we compute the normalized magnitude displacement

$$\delta_i = \frac{m_i^{\rm adv} - m_i}{e_i}, \tag{A.1}$$

which expresses the change of magnitude in units of the reported photometric uncertainty. The hard projection step of Section 3.1.2 enforces $|\delta_i| \leq k$ at every point by construction, and we use the empirical distribution of $|\delta_i|$ as a diagnostic of the actual perturbation strength reached by the PGD attack in each survey.

Table A.2 reports summary statistics of $|\delta_i|$ for the three surveys, computed over all observations of all light curves in the adversarial test sets. No observation violates the bound $k = 0.5$ in any of the surveys, as expected from the projection step. The fact that the median, the 95th percentile and the maximum of $|\delta_i|$ all reach the bound shows that the attack saturates the budget at well over half of the observations rather than producing trivially small perturbations. The mean ratio $|\delta_i|/k$ ranges from 0.91 in OGLE-III to 0.96 in *Gaia*, confirming that the attacks use most of the available pointwise budget while remaining inside the photometric box.

Appendix B: Per-class analysis

The global accuracy figures of Table 1 hide important class-level structure that is worth examining. We report the per-class accuracy of the RRE classifier on all three surveys, focusing on OGLE-III, where the largest baseline degradation occurs. Per-class accuracies are given in Table B.1.

Table B.2 reports the per-class F_1 of RRE with bootstrap CIs and explains the gap between accuracy and the class-balanced metrics in Table C.1. Robustness is uniform across the well-populated classes—every majority-class per-class F_1 moves by less than a point under attack—but the minority classes dominate the macro-F_1, which is the unweighted mean of the per-class F_1 values. On WISE the model never assigns the δ Scuti label ($n_{\rm test} = 220$), so its per-class F_1 is zero both clean and adversarial; together with the harder Cepheid class (per-class $F_1 \approx 0.55$) this pulls the WISE macro-F_1 down to 0.62 and balanced accuracy to 0.61, even though overall accuracy is 0.93. We state this explicitly rather than leaning on the aggregate number: the robustness we claim is a statement about *stability under perturbation*, and it holds class-by-class (no class loses more than ∼2 pp of per-class F_1), independently of the pre-existing class-imbalance limitation.

B.1. OGLE-III

On OGLE-III, the CNN baseline experiences per-class accuracy drops of −35.6 pp (ECL), −57.6 pp (RRLYR), −18.2 pp (CEP), and −3.5 pp (LPV) under attack. Every class except LPV is severely affected, with RRLYR in particular collapsing to 0.400 adversarial accuracy — well below its clean level and approaching random guess for a four-class problem. The largest absolute robustness gap is therefore on the class most relevant for cosmological distance measurements.

The RRE keeps all four OGLE classes stable: the four per-class drops fall within −0.9 to −0.0 pp, well within the binomial-sampling noise of the test split for the corresponding class size. The lowest clean accuracy of the RRE on OGLE is 0.957 on CEP — far above random guess — and even on this hardest class the RRE retains 0.951 adversarial accuracy. The improvement of the RRE over the CNN is therefore not driven by any single dominant class but by simultaneous robustness across all four.

B.2. Gaia DR2

On *Gaia* DR2, the RRE per-class accuracies are 0.954 (RRLYR), 0.947 (LPV), 0.802 (CEP), and 0.839 (DSCT). All four classes are stable under attack, with per-class drops within −0.8 pp. The lower absolute accuracy on CEP reflects the difficulty of distinguishing classical Cepheids from RR Lyrae and δ Scuti pulsators when only a few tens of observations per light curve are available. The qualitative robustness pattern is the same as on OGLE-III: whatever accuracy the model attains on the clean test set, it preserves under the adversarial attack.

Table A.1: Detectability of bounded adversarial perturbations at full perturbation fraction ($f = 1.0$): two-sample KS statistic (p-value in parentheses) between the clean-test and adversarial-test distributions of standard variability features. The OGLE attacks use $k = 1$; the *Gaia* attack uses $k = 0.5$. Scale-sensitive features (magnitude scatter σ, robust amplitude, Stetson K) do not separate the two samples, while order-sensitive statistics (von Neumann ratio $\eta_{\rm vN}$, lag-one autocorrelation ACF_1) detect the attack at high significance.

Survey	Surrogate	σ	Amp.	K	$\eta_{\rm vN}$	ACF_1
OGLE	CNN	0.020 (0.15)	0.022 (0.09)	0.019 (0.22)	0.092 (10^{-23})	0.092 (10^{-23})
OGLE	RNN	0.017 (0.34)	0.018 (0.26)	0.018 (0.24)	0.093 (10^{-24})	0.095 (10^{-25})
OGLE	RF surrogate	0.017 (0.33)	0.019 (0.19)	0.021 (0.14)	0.071 (10^{-14})	0.071 (10^{-14})
Gaia	CNN	0.019 (1.00)	0.013 (1.00)	0.019 (1.00)	0.013 (1.00)	0.013 (1.00)

Table A.2: Summary of the normalized adversarial displacement $|\delta_i| = |m_i^{\rm adv} - m_i|/e_i$ for the three surveys, aggregated over all observations of all light curves in the corresponding adversarial test set. The bound $k = 0.5$ is the per-point photometric-error budget defined in Section 3.1.2; no observation violates it in any of the surveys.

		$\lvert\delta_i\rvert$			
Survey	$n_{\rm obs}$	Mean	Median	P95	Max
OGLE-III	6,281,886	0.455	0.500	0.500	0.500
Gaia DR2	294,382	0.478	0.500	0.500	0.500
WISE	528,998	0.468	0.500	0.500	0.500

Table B.1: Per-class accuracy of the proposed RRE classifier on the three surveys, under the CNN-targeted PGD attack of Section 5.2 with $k = 0.5$. Drop = $\mathrm{acc_a} - \mathrm{acc_c}$ in percentage points: negative values indicate accuracy loss under attack; small positive values reflect statistical noise of the test split rather than a genuine improvement.

Survey	Class	$\mathrm{acc_c}$	$\mathrm{acc_a}$	drop (pp)
OGLE-III (RRE)	ECL	0.977	0.968	−0.9
	RRLYR	0.985	0.983	−0.2
	CEP	0.957	0.951	−0.6
	LPV	0.998	0.998	−0.0
Gaia DR2 (RRE)	RRLYR	0.954	0.948	−0.6
	LPV	0.947	0.944	−0.3
	CEP	0.802	0.797	−0.5
	DSCT	0.839	0.830	−0.9
WISE (RRE)	RRLYR	0.985	0.981	−0.4
	LPV	0.986	0.983	−0.3
	CEP	0.471	0.448	−2.3
	DSCT	0.000	0.000	0.0

Table B.2: Per-class F_1 of RRE (clean and adversarial) with 95% bootstrap CIs. The minority classes drive the gap between accuracy and macro-F_1: for WISE δ Scuti ($n_{\rm test} = 220$) the model never produces a correct prediction ($F_1 = 0$), which the aggregate accuracy masks. All values are percentages.

Survey	Class	F_1 clean	F_1 adv.
OGLE-III	ECL	98.1(97.8 – 98.4)	97.5(97.2 – 97.9)
	RRL	98.2(97.9 – 98.5)	97.8(97.5 – 98.2)
	CEP	96.9(96.3 – 97.6)	96.4(95.7 – 97.1)
	LPV	99.1(98.9 – 99.3)	99.0(98.7 – 99.2)
Gaia DR2	RRL	92.7(92.1 – 93.3)	92.1(91.5 – 92.7)
	LPV	94.5(93.9 – 95.0)	94.2(93.7 – 94.7)
	CEP	82.9(81.5 – 84.2)	82.4(80.9 – 83.7)
	DSCT	87.3(86.1 – 88.5)	86.3(85.0 – 87.4)
WISE	RRL	94.5(94.0 – 95.0)	94.1(93.6 – 94.6)
	LPV	98.0(97.7 – 98.3)	97.7(97.4 – 98.0)
	CEP	56.6(52.6 – 60.8)	54.4(50.2 – 58.3)
	DSCT	0.0(0.0 – 0.0)	0.0(0.0 – 0.0)

B.3. WISE

The WISE per-class accuracies remain highly asymmetric: 0.985 on rrlyr and 0.986 on lpv, but only 0.471 on cep, while none of the 220 dsct test curves is correctly classified: the model never predicts the dsct label, and 206 of them (94%) are assigned to rrlyr. This confusion is astrophysically plausible, since both classes are short-period pulsators with small mid-infrared amplitudes that the sparse WISE sampling poorly resolves, and the training set contains 18 times more rrlyr than dsct curves (Table H.2). The two well-represented classes are individually stable under attack (per-class drops of −0.3 pp each), and the attack leaves the dsct confusion unchanged (207 of 220 assigned to rrlyr). The limitation on the minority classes is therefore one of clean trainability, not adversarial vulnerability.

This pattern is explained by the class distribution of the underlying catalog (Table H.3). In WISE, RRLYR and LPV together contribute more than 84,000 light curves to the raw catalog, whereas CEP and DSCT together provide fewer than 3,500. Even after the per-class subsample described in Section 4, the WISE CEP and DSCT populations are essentially capped at their (small) raw catalog sizes, which is insufficient to train an end-to-end deep classifier to discriminate them from the abundant LPV class. We emphasize that this is a property of the WISE *catalog*, not of the proposed methodology: the same imbalance affects every classifier evaluated in Section 5.2, including the supervised baselines.

The standard imbalance remedy does not rescue the minority classes. As a diagnostic, we retrained the clean raw encoder (the V1 variant of Sect. 5.5) on WISE with balanced oversampling, resampling the minority classes with replacement up to the majority-class count (cf. Hosenie et al. 2020). The remedy con-

verts the failure mode rather than fixing it: the DSCT recall rises from zero to 0.46, but at a precision below 0.09 ($F_1 = 0.15$), i.e. the classifier now produces many low-precision DSCT labels; CEP behaves similarly ($F_1 = 0.43$). The cost elsewhere is severe — overall accuracy falls from 0.934 to 0.778 and the RRLYR per-class accuracy from 0.985 to 0.672 — and the macro-F_1 *decreases* from 0.62 to 0.58. With 878 DSCT training curves against 16,000 for LPV, and pulsation amplitudes that shrink toward the mid-infrared until they become comparable to the $W1$ photometric uncertainties, the class is not learnable at useful quality from this catalog by any remedy we tested. We therefore retain the four-class experiment with the limitation stated explicitly — balanced accuracy is reported alongside accuracy throughout, including the abstract — rather than replacing it with a "remedied" classifier that is worse on every aggregate metric.

Appendix C: Bootstrap confidence intervals

Tables C.1 and C.2 attach 95% confidence intervals to these numbers. On the paired bootstrap the clean→adversarial accuracy drop of RRE is only +0.4 pp [+0.3, +0.5] on OGLE-III, +0.5 pp [+0.4, +0.7] on *Gaia*, and +0.4 pp [+0.3, +0.6] on WISE; each interval excludes zero, yet the drops are a small fraction of the corresponding CNN drops (+30.6, +1.7 and +9.3 pp, i.e. between ∼3× on *Gaia* and ∼75× on OGLE-III larger). The adversarial-accuracy advantage of RRE over the CNN baseline is likewise unambiguous—+32.1 pp [+31.3, +32.9] on OGLE-III, +5.6 pp [+5.0, +6.1] on *Gaia* and +10.7 pp [+10.0, +11.3] on WISE—so none of the headline robustness gaps is an artifact of test-set sampling.

Appendix D: Model-specific attacks at both budgets

Table D.1 reports these baseline stress tests at the default ($k = 0.5$) and maximal ($k = 1.0$) photometric budgets, complementing the transferred-attack results of the main text.

Appendix E: Method details

E.1. Latent-space stability under perturbation

The accuracy figures of Sections 5.3–B show that the predicted class of the RRE classifier is essentially unchanged by the adversarial attack, but they do not by themselves explain how this stability arises inside the model. In this subsection we examine the internal representation of clean and adversarial light curves to identify the mechanism by which the RRE produces stable predictions.

We compare the learned latent space of the full RRE with that of its V1 ablation (Section 5.5), trained on the same raw two-tower $(\Delta t, \Delta m, \sigma)$ representation and the same OGLE-III training split with clean cross-entropy only and no adversarial component. The two encoders share the same Conv1D backbone, the same latent dimensionality, and the same optimization protocol; they differ solely in the training objective, so any difference between the two latent spaces is attributable to the adversarial and regularization terms of the RRE loss rather than to the architecture.

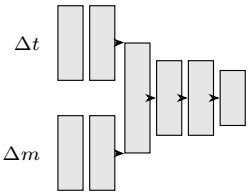
Figure E.2 shows kernel-density estimates of the first three latent means (μ_1, μ_2, μ_3) of the two encoders, for a representative class (LPV) of OGLE-III, separating the clean test light curves (blue) from their adversarial counterparts (orange). Panels (a)–(c) correspond to the V1 ablation and panels (d)–(f) to the full RRE.

For the V1 ablation (left column), the clean and adversarial distributions are visibly shifted along the latent axes: although the perturbation moves every observation by at most $0.5\, e_i$ in magnitude, an encoder trained without any adversarial objective maps the perturbed input to a measurably different region of the latent space. Averaged over all 48 latent dimensions, the clean-to-adversarial displacement of the per-dimension mean is $0.51\,\sigma$ for V1, with a two-sample Kolmogorov–Smirnov statistic of 0.28 between the clean and adversarial marginals, confirming that the shift seen in panels (a)–(c) is representative of the full latent space. This is the latent-space footprint of the adversarial perturbation that drives the misclassification of the baseline classifiers in Section 5.2.

For the full RRE (right column), the clean and adversarial distributions are essentially superimposed in all three latent dimensions, and the all-dimension shift drops to $0.21\,\sigma$ (KS statistic 0.10), less than half that of the V1 ablation. The supervised contrastive component $\mathcal{L}_{\mathrm{SupCon}}$ of the V5 loss (Section 3.2.4) explicitly forces clean and adversarial views of the same light curve to be mapped to nearby points in latent space, while the VIB $\mathcal{L}_{\mathrm{VIB}}$ compresses the representation enough to absorb the non-informative variability introduced by the perturbation. The resulting latent space is intrinsically insensitive to perturbations within the photometric-error budget, which is the structural reason behind the near-zero adversarial drop reported in Tables 1 and B.1.

1. Adversarial Examples Generation

1.1 Surrogate two-tower CNN training



1.2 Photometrically bounded PGD attack

- Observed light curve $\{(t_i, m_i, e_i)\}$
- PGD on the Δm channel (T=20 steps, smoothness penalty $\lambda_{\text{sm}}=10^{-3}$)
- Final hard projection in raw magnitudes: $m_i^{\text{adv}} \leftarrow \text{clip}\big(m_i^{\text{adv}},\, m_i - k\,e_i,\, m_i + k\,e_i\big)$
- Pointwise budget $|m_i^{\text{adv}} - m_i| \le k\,e_i$, k=0.5
- Adversarial light curve saved to disk

1.3 Transferability property

$$\{x_{\text{adv}}\} \in \mathbb{R}^n \xrightarrow{\text{RF}} P(\hat{y} \mid x_{\text{adv}})$$
$$\{x_{\text{adv}}\} \in \mathbb{R}^n \xrightarrow{\text{CNN}} P(\hat{y} \mid x_{\text{adv}})$$
$$\{x_{\text{adv}}\} \in \mathbb{R}^n \xrightarrow{\text{RNN}} P(\hat{y} \mid x_{\text{adv}})$$

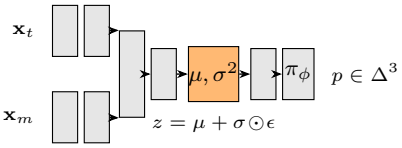
adv. light curves

2. Robust Raw Encoder (RRE) with adversarial training

2.1 Raw differential representation (no period, no folding)

- $\Delta t_i = t_i - t_{i-1}, \quad \Delta m_i = m_i - m_{i-1}$
- $\sigma_{\Delta m_i} = \sqrt{e_i^2 + e_{i-1}^2}$ (propagated uncertainty)
- Uniformly subsample/pad to $L = 500$ epochs; global per-channel standardization (train-split statistics)
- Two-tower input: $\mathbf{x}_t = \Delta t \in \mathbb{R}^{L\times 1}$, $\mathbf{x}_m = [\Delta m, \sigma_{\Delta m}] \in \mathbb{R}^{L\times 2}$

2.2 Two-tower encoder with variational bottleneck



Per-channel Conv1D towers + joint conv block + GAP, latent $D = 48$

2.3 Inner PGD attack in raw-magnitude space

$\epsilon_i \leftarrow k\,e_i \quad \delta \sim \mathcal{U}(-\epsilon, \epsilon), \;\; \delta_1$=0 (anchor)
for $s = 1, \ldots, S = 3$ **do**
  $\mathbf{x}' \leftarrow$ recompute Δm from $m+\delta$ (differencing)
  $\delta \leftarrow \delta + 0.4\,\epsilon\,\text{sign}\,\nabla_\delta\,\text{CE}\big(y,\; \pi_\phi(\mu_\theta(\mathbf{x}'))\big)$
  $\delta \leftarrow \text{clip}(\delta, -\epsilon, +\epsilon) \quad \Rightarrow |m_i^{\text{adv}} - m_i| \le k e_i$ always
return $\mathbf{x}_{\text{adv}}$

2.4 Combined training objective

$$\mathcal{L} = \underbrace{\text{CE}(y,p)}_{\text{clean}} + \lambda_{\text{adv}} \underbrace{\text{CE}(y,p_{\text{adv}})}_{\text{adv-CE}} + \lambda_{\text{tr}} \underbrace{\text{KL}(p \,\|\, p_{\text{adv}})}_{\text{TRADES}} + \lambda_{\text{sc}} \underbrace{\mathcal{L}_{\text{SupCon}}(z, z_{\text{adv}}, y)}_{\text{latent consistency}} + \beta \underbrace{\text{KL}\big(q_\theta(z|x) \,\|\, \mathcal{N}(0,I)\big)}_{\text{VIB compression}}$$

one shared $\mathbf{x}_{\text{adv}}$ feeds all adversarial terms;
$(\lambda_{\text{adv}}, \lambda_{\text{tr}}, \lambda_{\text{sc}}, \beta) = (1,\ 2,\ 0.5,\ 5\times 10^{-4})$

2.5 Robust classifier evaluation

$$\{x_{\text{adv}}\} \in \mathbb{R}^n \xrightarrow{\text{RF}} P(\hat{y} \mid x_{\text{adv}})$$
$$\{x_{\text{adv}}\} \in \mathbb{R}^n \xrightarrow{\text{CNN}} P(\hat{y} \mid x_{\text{adv}})$$
$$\{x_{\text{adv}}\} \in \mathbb{R}^n \xrightarrow{\text{RNN}} P(\hat{y} \mid x_{\text{adv}})$$
$$\{x_{\text{adv}}\} \in \mathbb{R}^n \xrightarrow{\textbf{ours (RRE)}} P(\hat{y} \mid x_{\text{adv}})$$

+ adaptive white-box PGD on RRE itself
(raw-magnitude budget, 40 iter., 5 random restarts)

Fig. E.1: Overview of the two-stage framework. **Stage 1 (left): photometry-constrained adversarial generation.** A surrogate two-tower CNN is trained on the differential representation $(\Delta t, \Delta m)$ of each light curve. Adversarial examples are crafted with a T=20-step PGD attack that perturbs only the magnitudes, under the per-point photometric budget $|m_i^{\text{adv}} - m_i| \le k\,e_i$ with k=0.5; the resulting curves are transferred to the RF, CNN and RNN baselines. **Stage 2 (centre and right): Robust Raw Encoder (RRE).** The encoder ingests the raw differential representation $(\Delta t, \Delta m, \sigma_{\Delta m})$ directly. Two 1D-convolutional towers process the time and magnitude channels; their features are concatenated and compressed through a VIB, with the deterministic mean μ used at inference. An inner PGD loop perturbs the raw magnitudes within the pointwise budget $|\delta_i| \le k\,e_i$ (k=0.5, 3 steps, magnitude displacement clipped at every iteration) and recomputes Δm by differencing, so that clean and on-the-fly adversarial views pass through the same encoder during training. A classifier head and an ℓ_2-normalized projection head (for supervised contrastive learning) feed the full training objective $\mathcal{L} = \mathcal{L}_{\text{CE}}(\text{clean}) + \lambda_{\text{adv}}\mathcal{L}_{\text{CE}}(\text{adv}) + \lambda_{\text{tr}}\mathcal{L}_{\text{TRADES}} + \lambda_{\text{sc}}\mathcal{L}_{\text{SupCon}} + \beta\,\mathcal{L}_{\text{VIB}}$ (λ_{adv}=1.0, λ_{tr}=2.0, λ_{sc}=0.5, β=5×10^{-4}). The trained model is evaluated under the same transferred attacks as the baselines.

Table C.1: Clean and adversarial test performance with 95% bootstrap confidence intervals ($B = 2000$). Adversarial inputs use the photometric budget $k\,e_i$ ($k = 0.5$); the attack optimizes bounded raw-magnitude displacements and recomputes Δm. Intervals are percentile bootstrap over the test set; the clean→adv drop uses a paired bootstrap. All values are percentages.

		Accuracy		Balanced acc.		Macro-F_1	
Survey	Model	Clean	Adv.	Clean	Adv.	Clean	Adv.
OGLE-III	CNN	96.4(96.1 – 96.7)	65.8(65.0 – 66.7)	94.0(93.5 – 94.5)	65.3(64.5 – 66.1)	94.9(94.5 – 95.4)	62.9(62.1 – 63.6)
	RRE	98.3(98.1 – 98.5)	97.9(97.7 – 98.2)	97.9(97.6 – 98.2)	97.5(97.2 – 97.8)	98.1(97.8 – 98.4)	97.7(97.4 – 98.0)
Gaia DR2	CNN	86.8(86.2 – 87.4)	85.0(84.4 – 85.7)	82.5(81.8 – 83.3)	80.3(79.6 – 81.1)	83.5(82.7 – 84.2)	81.3(80.5 – 82.1)
	RRE	91.1(90.6 – 91.6)	90.6(90.0 – 91.1)	88.5(87.8 – 89.2)	88.0(87.3 – 88.6)	89.3(88.7 – 90.0)	88.7(88.1 – 89.4)
WISE	CNN	91.4(90.8 – 91.9)	82.1(81.3 – 82.8)	53.6(52.7 – 54.4)	45.2(44.7 – 45.7)	54.2(53.0 – 55.4)	44.6(44.1 – 45.1)
	RRE	93.2(92.6 – 93.7)	92.8(92.2 – 93.3)	61.0(59.9 – 62.2)	60.3(59.2 – 61.4)	62.3(61.2 – 63.4)	61.5(60.4 – 62.6)

Table C.2: Paired bootstrap 95% CIs for the accuracy quantities that the robustness claims rest on. The clean→adv *drop* and the RRE−CNN *gap* are computed within each bootstrap replicate (paired), so the intervals account for the correlation between estimators. All quantities are in percentage points; none of the intervals contains zero.

Survey	Quantity (accuracy)	Value (pp), 95% CI
OGLE-III	CNN clean−adv drop	+30.6 [+29.8, +31.4]
	RRE clean−adv drop	+0.4 [+0.3, +0.5]
	RRE − CNN (clean)	+1.9 [+1.6, +2.2]
	RRE − CNN (adv)	+32.1 [+31.3, +32.9]
Gaia DR2	CNN clean−adv drop	+1.7 [+1.5, +2.0]
	RRE clean−adv drop	+0.5 [+0.4, +0.7]
	RRE − CNN (clean)	+4.3 [+3.8, +4.9]
	RRE − CNN (adv)	+5.6 [+5.0, +6.1]
WISE	CNN clean−adv drop	+9.3 [+8.7, +9.9]
	RRE clean−adv drop	+0.4 [+0.3, +0.6]
	RRE − CNN (clean)	+1.8 [+1.4, +2.2]
	RRE − CNN (adv)	+10.7 [+10.0, +11.3]

Table D.1: OGLE-III accuracy of each baseline under the same photometric threat model, evaluated at two per-point budgets: the default $k = 0.5$ used throughout the paper and the full one-reported-error budget $k = 1.0$ (a displacement of one full reported uncertainty per point). The numerical search is PGD for the CNN/RNN and Square Attack (Andriushchenko et al. 2020) for the RFs. RF-FATS16 is the 16-feature RF of Table 1, attacked by perturbing the raw magnitudes of the full light curve and recomputing all features at every query (stratified subsample of 800 test curves, ~550 queries to plateau); RF-$\Delta m7$ is the simplified 7-feature variant in standardized Δm space (4,000 queries, full test set). Cells marked † (the RNN and RF-$\Delta m7$ rows and the CNN $k = 1.0$ entry) were obtained on independently trained instances of the corresponding baselines, whose clean accuracies differ by at most 0.5 pp from those of Table 1. Attack labels are abbreviated WB (white-box) and BB (black-box). Drop = $\mathrm{acc_a} - \mathrm{acc_c}$ in percentage points.

			$k = 0.5$		$k = 1.0$	
Model	Attack	$\mathrm{acc_c}$	$\mathrm{acc_a}$	drop	$\mathrm{acc_a}$	drop
CNN	PGD (WB)	0.964	0.658	−30.6	0.465†	−49.9
RNN†	PGD (WB)	0.950	0.592	−35.8	0.427	−52.4
RF-FATS16	Square (BB)	0.969	0.904	−6.5	0.805	−16.4
RF-$\Delta m7^\dagger$	Square (BB)	0.944	0.818	−12.6	0.724	−22.0

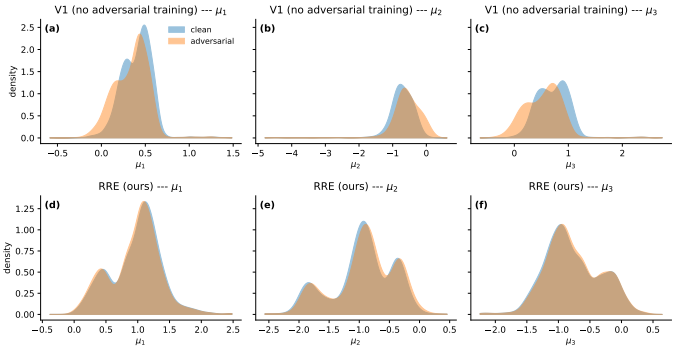


Fig. E.2: Kernel-density estimates of the first three latent means (μ_1, μ_2, μ_3) for OGLE-III LPV light curves, clean (blue) and adversarial (orange). Panels (a)–(c) (left column) correspond to the V1 ablation (clean cross-entropy only, no adversarial component); panels (d)–(f) (right column) correspond to the proposed full RRE classifier. Both encoders share the same raw two-tower $(\Delta t, \Delta m, \sigma)$ architecture and differ solely in the training objective. The clean and adversarial distributions are visibly shifted in the V1 latent space and superimposed in the RRE latent space, illustrating the role of the adversarial, supervised contrastive, and information-bottleneck components of the full V5 objective.

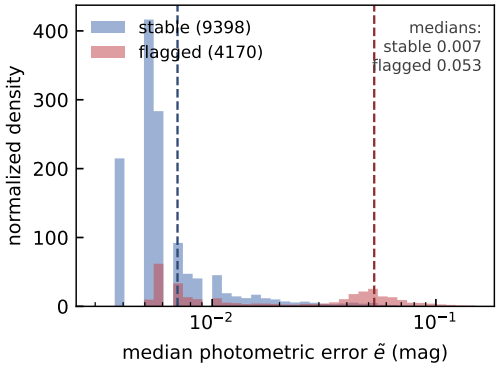


Fig. E.3: Distribution of the median reported photometric error $\tilde{e}$ for OGLE-III sources whose CNN prediction is flipped by the transferred attack (flagged) versus those that are stable. Flagged sources are systematically noisier (median 0.053 vs. 0.007 mag; Mann–Whitney $p \ll 10^{-100}$), showing that the robustness flag of Section 6 tracks genuine data quality. Dashed lines mark the two medians; the horizontal axis is logarithmic.

Table E.1: Per-source robustness flag: fraction of test predictions that change under the transferred attack ($k = 0.5$), i.e. the sources that the audit would route to follow-up. Absolute counts in parentheses. The margin-resolved view is given in Fig. F.1.

Survey	CNN	RRE (ours)
OGLE-III	30.7% (4170/13568)	0.4% (56/13568)
Gaia DR2	1.8% (209/11483)	0.5% (63/11483)
WISE	9.8% (853/8704)	0.5% (45/8704)

Appendix F: Additional robustness controls

This appendix collects the control experiments summarized in Sect. 5.5: the random-noise-augmentation baseline, the component controls without adversarial cross-entropy, and the VIB-only β-sweep.

F.1. Component controls without adversarial cross-entropy

To separate adversarial cross-entropy from the other terms, we trained controls that activate each family of terms without AdvCE (Table F.1).

Table F.1: Component controls on OGLE-III under the transferred CNN-targeted attack ($k = 0.5$). These runs use the same raw two-tower architecture and training split as Table 2, but isolate whether a loss family can produce robustness without adversarial cross-entropy. Drop is $\mathrm{acc_c} - \mathrm{acc_a}$ in percentage points, so smaller is better.

Variant	Active components	$\mathrm{acc_c}$	$\mathrm{acc_a}$	drop
V2	CE + AdvCE	0.984	0.979	0.45
TRADES-only	CE + TRADES	0.983	0.979	0.37
Aux-no-AdvCE	CE + TRADES + SupCon + VIB	0.982	0.977	0.49
SupCon-only	CE + SupCon	0.988	0.943	4.44
VIB-only	CE + VIB	0.985	0.905	8.01

The controls refine the interpretation of the V5 objective. Clean cross-entropy plus TRADES alone reaches 0.983 clean and 0.979 adversarial accuracy, indistinguishable from V2 under the transferred attack; the combination of TRADES, SupCon and VIB without adversarial cross-entropy remains similarly robust ($0.982 \rightarrow 0.977$). By contrast, SupCon alone is only partially robust ($0.988 \rightarrow 0.943$), and VIB alone is not robust (Sect. F.2). The pairwise error-overlap analysis also does not support a simple interpretation in which each term protects a separate observational regime: relative to V2, V4 rescues 78 of the 280 V2 adversarial errors while introducing 46 new ones, whereas V5 and TRADES-only exchange nearly equal numbers of objects. The empirical message of the ablation is that the raw representation becomes robust when clean and photometrically bounded adversarial views are explicitly coupled during training; the full V5 objective keeps this robust solution while also imposing the representation-level structure that motivates the latent analysis.

F.2. The bottleneck alone provides no robustness

Because the variational bottleneck has occasionally been proposed as an intrinsically robustness-inducing mechanism (Alemi et al. 2017), we also trained the complementary configuration: clean cross-entropy plus VIB only, with *no* adversarial component, sweeping $\beta \in \{5 \times 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}\}$. All four models retain high clean accuracy (0.985–0.987) but lose 8.0–9.4 pp under the same transferred attack — a drop that *grows* with the compression strength — and a stronger bottleneck ($\beta = 1$) collapses to the majority class. Information compression alone therefore does not confer robustness within the photometric threat model: the explicit clean–adversarial coupling provided by AdvCE or TRADES is the operative ingredient, and VIB earns its place only as a latent-space regularizer on top of that coupling. This finding independently corroborates Korshunova et al. (2021), who showed on image benchmarks that information-bottleneck models alone are not a strong defense and that earlier positive results were affected by gradient obfuscation — a concern our adaptive evaluation (Sect. 5.7) addresses by attacking through the deterministic encoder mean.

Appendix G: Gradient-free, margin-based, and attack-potency checks: protocols

Table G.1: Adaptive white-box PGD evaluated on RRE's *own* gradient ($T = 40$ steps, per-point step $0.1\,\epsilon_i$, 5 random restarts, raw-magnitude displacement bounded by $k\,e_i$ with $k = 0.5$), following the adaptive-evaluation recommendations of Carlini et al. (2019). Even under a first-order attack that directly follows the defended model's gradient, the accuracy drop does not exceed one percentage point. This is evidence against simple gradient masking, not a certificate of robustness. All values are percentages.

Survey	Clean	Adaptive PGD	Drop (pp)
OGLE-III	98.3	97.3	−1.0
Gaia DR2	91.1	90.4	−0.7
WISE	93.2	92.5	−0.7

Following the evaluation checklist of Carlini et al. (2019), we complement the adaptive PGD with three further checks on OGLE-III.

First, a *gradient-free* attack: because the encoder contains a stochastic bottleneck — precisely the component that Korshunova et al. (2021) showed can obfuscate gradients — we ran the same score-based Square-style search used against the RF (Sect. 5.2), perturbing the raw magnitudes of a stratified subsample of 800 test curves within the $k = 0.5$ budget and querying the RRE class probabilities at every step. After 1,000 iterations the accuracy declines only from 0.976 to 0.974 (−0.25 pp), against −0.37 pp for the white-box adaptive PGD run on the *same* 800-curve subsample ($0.976 \rightarrow 0.973$; the full-test white-box drop is −1.0 pp): the gradient-based attack is consistently stronger than the gradient-free one, which is the expected signature of a model whose gradients are informative rather than masked (Athalye et al. 2018).

Second, a *margin-based* objective: we replaced the cross-entropy of the adaptive attack with the untargeted DLR loss of APGD (Croce & Hein 2020), computed exactly from the classifier log-probabilities (softmax shift-invariance makes logit differences equal to log-probability differences). At $T = 40$ with five restarts the DLR attack reaches 0.973, identical to the cross-entropy attack, and extending to $T = 100$ changes nothing (0.973): the attack is converged, and the margin objective finds no additional adversarial examples.

Third, an *attack-potency* diagnostic outside the threat model: rerunning the adaptive PGD with the budget inflated to $k = 5$ and $k = 20$ — physically inadmissible displacements of five and twenty reported uncertainties per point — collapses the accuracy to 0.135 (below the four-class random-guess level) and 0.007, respectively. The same attack configuration that annihilates the

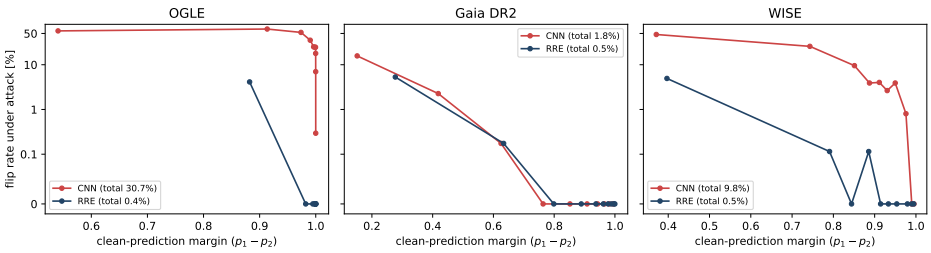


Fig. F.1: The per-source robustness flag in practice (discussed in Sect. 6): fraction of test predictions that change under the transferred attack ($k = 0.5$) as a function of the clean-prediction margin $p_1 - p_2$, in margin deciles (symlog vertical scale; legend gives the survey-wide flip fraction of each model). Deciles are computed per model and each point is drawn at the median margin of its decile, so the horizontal extent of a curve reflects that model's confidence distribution: the RRE curve on OGLE-III begins near margin 0.9 because even its lowest-confidence decile is that confident. For the defended RRE (blue), prediction changes are rare and concentrate at low margin, so margin plus the adversarial flip flag identifies the small subset of sources (∼0.5%) whose classifications warrant follow-up. For the undefended CNN (red), flips persist at margins close to 1 on OGLE-III and WISE: high reported confidence does not certify stability within the photometric error bars.

model when given room is therefore the one that fails to move it inside the photometric error bars, confirming that the robustness at admissible budgets reflects the threat model, not an under-converged attack.

Appendix H: Survey details

This appendix documents the composition of the three catalogs used in this work. Table H.1 lists the native variability classes of each survey together with the number of sources available before any selection, as compiled by Becker et al. (2020). The native labels differ between surveys — OGLE-III uses aggregate morphological classes, *Gaia* DR2 reports the pulsation-mode labels of its variability pipeline, and the WISE catalog inherits the OGLE-III nomenclature — so Table H.3 shows how they are grouped into the four working classes used throughout the paper, and Table H.2 gives the resulting per-class train/test counts after the per-class cap and the stratified split. Together they make explicit the class imbalance that drives the WISE per-class results of Appendix B: the CEP and DSCT populations there are one to two orders of magnitude smaller than RRLYR and LPV.

Table H.1: Native catalog labels of each survey, grouped into the working classes of this paper: RR Lyrae (RRL), Cepheids (CEP), long-period variables (LPV), δ Scuti/SX Phoenicis (DSCT), and eclipsing binaries (ECL). Numbers are sources available before any selection (Becker et al. 2020); bold rows give the class totals. Subtype labels use a common notation across surveys. OGLE-III provides only aggregate classes (its ECL class aggregates EC, ESD, and ED; LPV aggregates Mira, SRV, and OSARG; RRL aggregates RRab and RRc), so its subtype counts are not listed. A dash means that the class or subtype is absent from that catalog.

Class	Subtype	OGLE-III	*Gaia* DR2	WISE
RRL		**33,933**	**195,515**	**20,488**
	RRab		162,244	16,412
	RRc		32,331	3,831
	RRd		834	241
	anomalous RRd		106	4
CEP		**7,836**	**8,542**	**2,419**
	classical		6,488	1,884
	type II		1,740	482
	anomalous		314	53
LPV		**275,857**	**150,215**	**63,891**
	Mira + SRV		150,215[a]	10,001
	OSARG		–	53,890
DSCT		2,822[b]	**8,870**	**1,098**
ECL		**37,840**	–	–
Not used	non-variable	34,815	–	32,795
	NC, ELL, C[c]	–	–	2,548
Total		393,103	363,142	123,239

Notes. [a] *Gaia* DR2 reports Miras and semiregulars as a single class (MIRA_SR); in WISE they are 1,396 Miras and 8,605 SRVs. [b] Present in the OGLE-III catalog but not used: the OGLE-III experiment adopts its four most populated classes. [c] Non-classified, ellipsoidal, and other WISE groups.

Table H.2: Final per-class train/test counts and epoch distribution, from a stratified 70/10/20 split (seed 42) after capping each class at 20,000 objects. Train aggregates train+validation (70% + 10%); minority-class counts are exact to within rounding (≤ 4). Epochs are median [IQR] over the test set; $L = 500$ is the imposed maximum length (Sect. 3.2.1), not the native cadence.

Survey	Class	Train	Test	Epochs
OGLE-III	ECL	16,000	4,000	500
	RRL	16,000	4,000	500
	CEP	6,268	1,568	500
	LPV	16,000	4,000	500
	All	54,268	13,568	500 [436,500]
Gaia DR2	RRL	16,000	4,000	20
	LPV	16,000	4,000	23
	CEP	6,833	1,709	26
	DSCT	7,096	1,774	20
	All	45,929	11,483	23 [15,30]
WISE	RRL	16,000	4,000	29
	LPV	16,000	4,000	82
	CEP	1,935	484	49
	DSCT	878	220	27
	All	34,813	8,704	48 [28,84]

Table H.3: Per-survey class counts after grouping the native catalog labels of Table H.1 into the four working classes of this paper, before the per-class cap of 20,000 objects and the train–test split (Table H.2). Per-class numbers explain the WISE failure mode on CEP and DSCT discussed in Appendix B.

Survey	RRLYR	ECL	LPV	CEP	DSCT
OGLE-III	33,933	37,840	275,857	7,836	–
Gaia DR2	195,515	–	150,215	8,542	8,870
WISE	20,488	–	63,891	2,419	1,098

Appendix I: Baseline configurations

All baselines share the split, seed, and preprocessing of Section 5.1; the trained checkpoints and training scripts are part of the released repository (Section 3.3).

Random Forest (RF-FATS16). he RF operates on 16 handcrafted, period-independent features computed from the raw magnitudes and reported errors of each curve (Table I.1); they follow the spirit of the FATS library (Nun et al. 2015) but are recomputed by our pipeline, which is what allows the black-box attack of Sect. 5.2 to re-evaluate them at every query. The classifier is a `scikit-learn` `RandomForestClassifier` with 300 trees and maximum depth 25; all other hyperparameters are left at their `scikit-learn` defaults ($\sqrt{d}$ features per split, minimum leaf size 1), and the random seed is fixed to 42 for reproducibility. These values were set a priori rather than tuned by grid search: random-forest accuracy is known to be insensitive to them once a few hundred trees are used (Breiman 2001; Hastie et al. 2009), and the resulting classifier is a strong baseline (96.9% clean accuracy on OGLE-III), so its vulnerability is not an artifact of under-fitting.

Surrogate CNN. The two-tower CNN of Sect. 3.1.1 applies two weight-shared convolutional blocks to each channel — Conv1D(32, kernel 15, stride 2, ReLU) followed by Conv1D(16, kernel 11, stride 2, ReLU) — concatenates the two feature maps, and classifies through Flatten → Dropout(0.5) → Dense(64, tanh) → Dropout(0.5) → Softmax(4), for 2.6×10^5 trainable parameters. It is trained with Adam (learning rate 10^{-3}), categorical cross-entropy, batch size 256, for up to 50 epochs with early stopping on the validation loss (patience 5, best weights restored).

Table I.1: The 16 features of the RF baseline, computed from the raw magnitudes m_i and reported errors e_i (no period information).

Feature	Definition
`n_obs`	number of observations
`median`	median magnitude
`weighted_mean`	inverse-variance weighted mean
`weighted_std`	inverse-variance weighted std. dev.
`amplitude`	$(\max m - \min m)/2$
`iqr`	interquartile range
`mad`	median absolute deviation
`skew`, `kurtosis`	standardized third/fourth moments
`stetson_k`	Stetson K robust shape statistic
`beyond1std`	fraction of points beyond 1σ of mean
`slope_p10`, `slope_p90`	sorted-magnitude tail slopes
`eta`	von Neumann ratio η
`mean_err`, `median_err`	mean / median reported error

Recurrent baseline (GRU). The RNN concatenates the $(\Delta t, \Delta m)$ channels into a $(L, 2)$ sequence processed by a single GRU layer of 64 units followed by a softmax layer. It is trained with Adam (10^{-3}), categorical cross-entropy, batch size 256, early stopping on the validation accuracy (patience 4) and learning-rate halving on plateau (patience 2).